

ChiedzaAI

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Clarity for AI-driven transformation.

Module	Focus area	Purpose
ZivaBasa	Jobs, employment, productivity and skills	Forecasts how AI and digital transformation affect workforce structure, job displacement, job creation, productivity and skills demand.
ZivaDzidzo	Education	Forecasts how AI and digital transformation affect learning outcomes, teacher roles, student performance, curriculum needs and future skills.
ZivaBusiness	Revenue, profit and customers	Forecasts how AI and digital transformation affect company performance, customer behaviour, revenue growth, profitability and operating efficiency.
ZivaUpfumi	Economy, GDP and informal market	Forecasts how AI and digital transformation affect GDP, informal markets, productivity, employment, financial inclusion and sector growth.

Problem we solving

- 1. Unknown employment structure, jobs and skills disruption**
- 2. Absence of decision-support tools**
- 3. Limited datasets and evidence**

Significance of the Project

1. First explainable evidence base

Maps AI and DT impact on Zimbabwe's banking workforce.

2. Employment and skills insight

Supports analysis of workforce change in the digital economy.

3. Workforce planning support

Guides reskilling, upskilling, and human-capability planning.

4. Productivity and profitability value

Helps banks improve efficiency and manage HR-related costs.

5. Policy and regulatory decision support

Provides scenario analysis for employment, productivity, and skills planning.

Contributions

1. Explainable multi-task DL framework

Jointly predicts employment, productivity, and workforce skills impacts.

2. Causal-consistent XAI layer

Improves transparency, interpretation, and policy defensibility.

3. Privacy-preserving bank collaboration

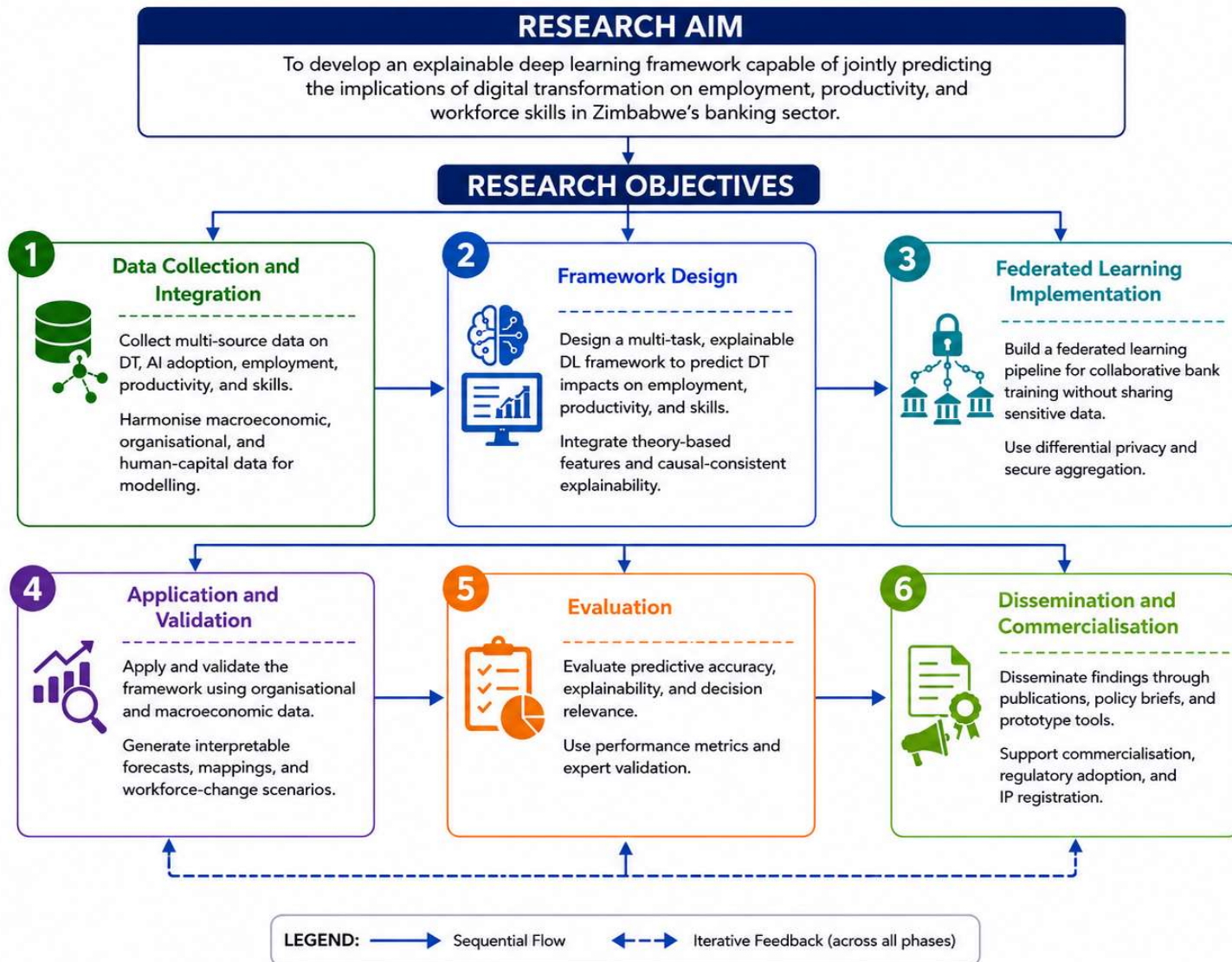
Enables federated learning without exposing sensitive data.

4. Four-layer policy-ready model

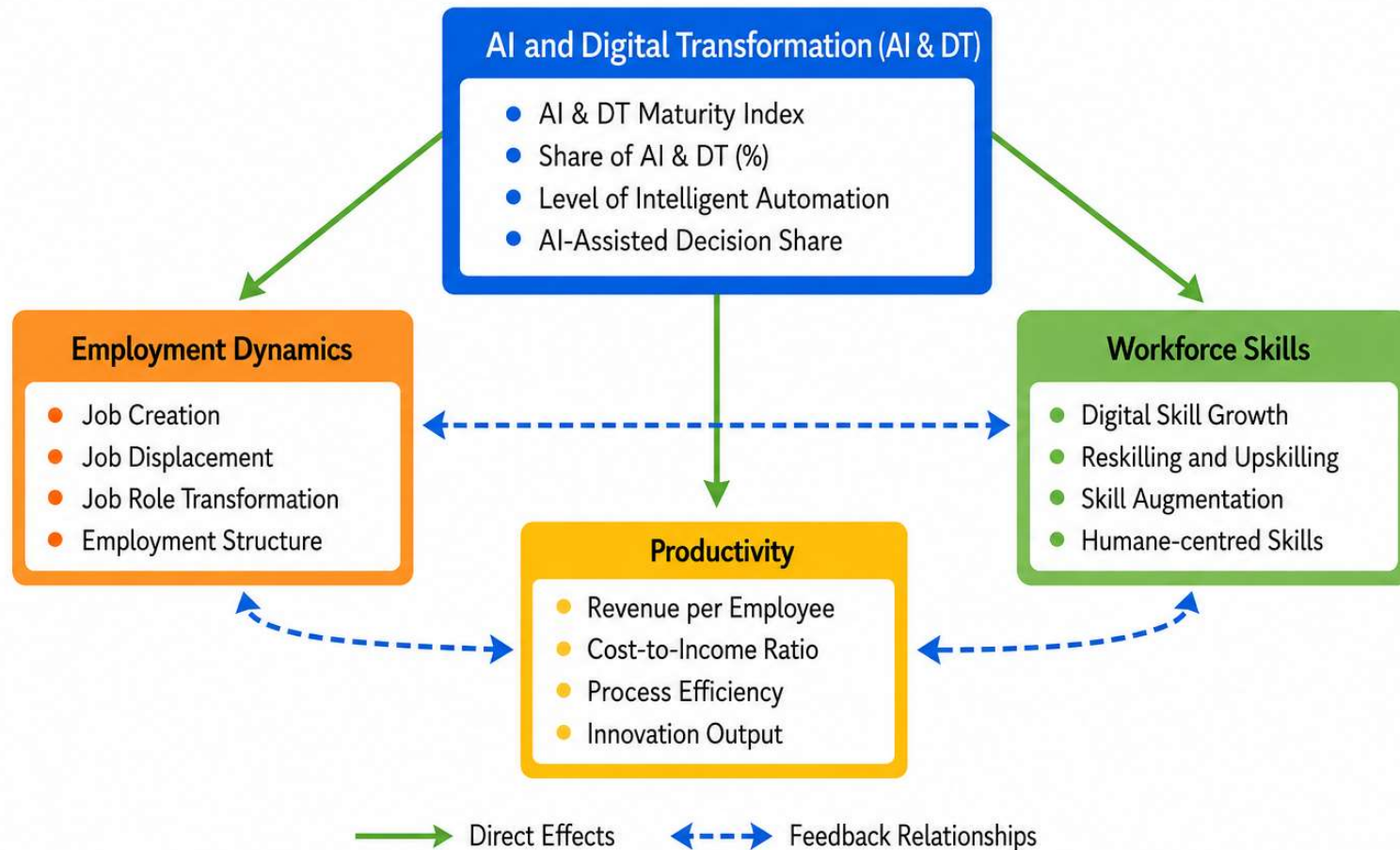
Links AI & Digital Transformation inputs, Deep Learning processing, XAI, and workforce outputs.

5. Decision-support toolkit

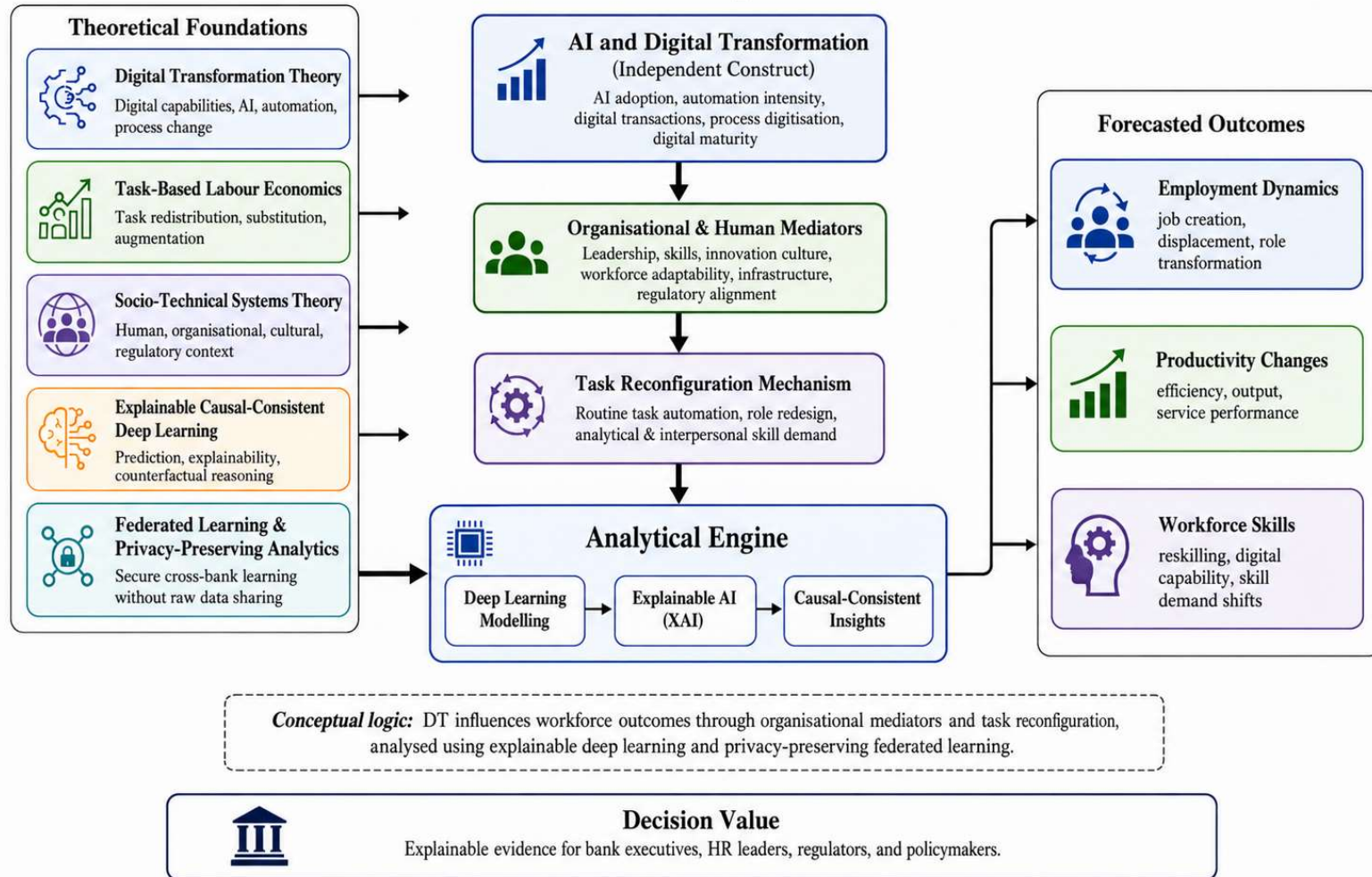
Supports reskilling, productivity planning, regulation, and responsible AI adoption.

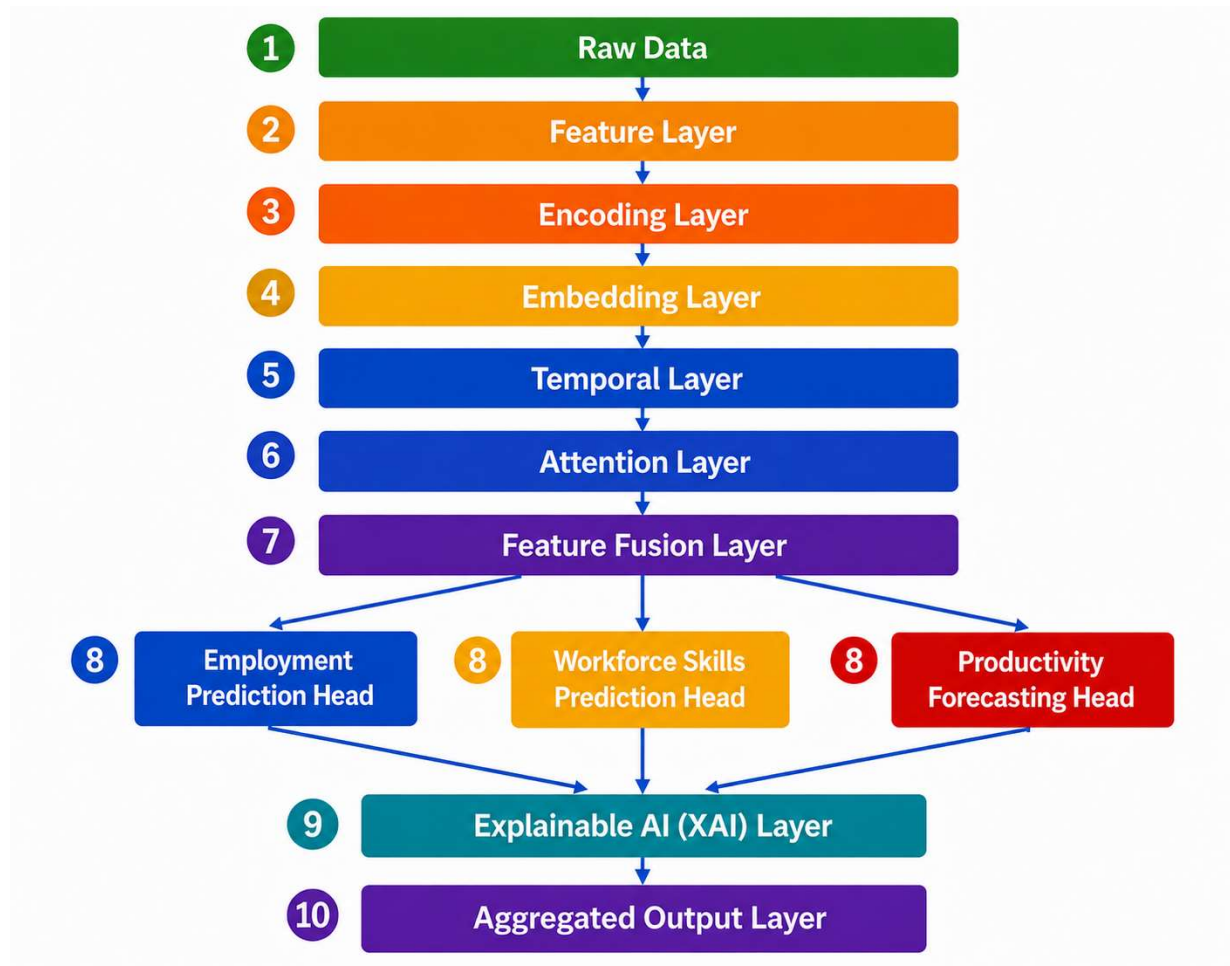


Conceptual Framework: Impacts of AI and Digital Transformation on Employment Dynamics, Workforce Skills, and Productivity in Zimbabwe's Banking Sector

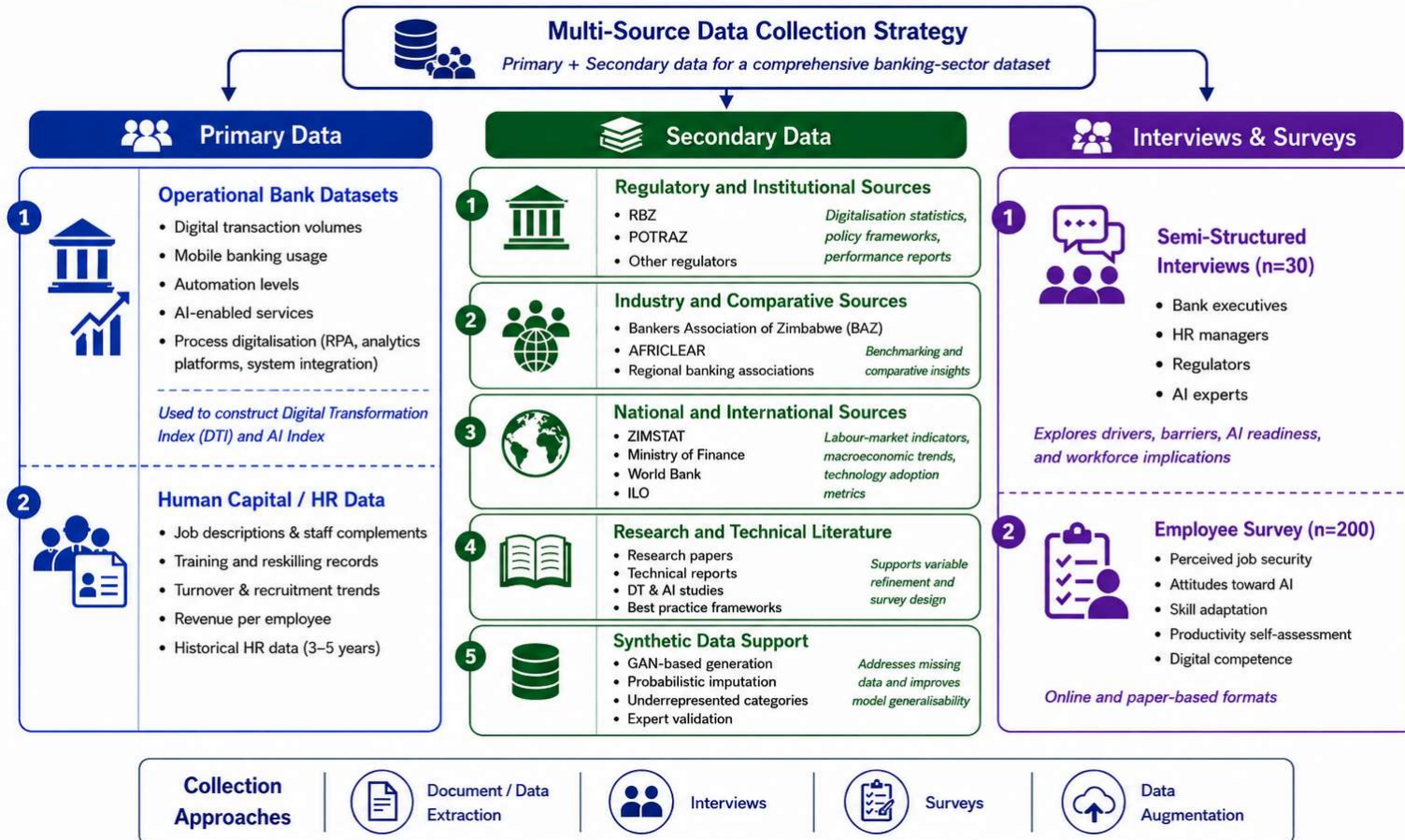


Theoretical and Conceptual Framework





Data Sources and Collection Approaches



DATA COLLECTION METHODS

DATA COLLECTION METHODS	
	Direct Corporate Collection From participating banks (HR, Operations, IT)
	Structured Interviews n = 30 with executives, HR managers, regulators, AI experts
	Employee Surveys n = 200 across banks (Online & Paper-based)
	Public Data Acquisition From regulators, government agencies & global institutions
	Literature & Report Review Systematic search & coding of relevant studies
	Synthetic Data Generation & Imputation GANs, Probabilistic Models, Expert Validation

Criteria for Determining the Number of Survey Questions

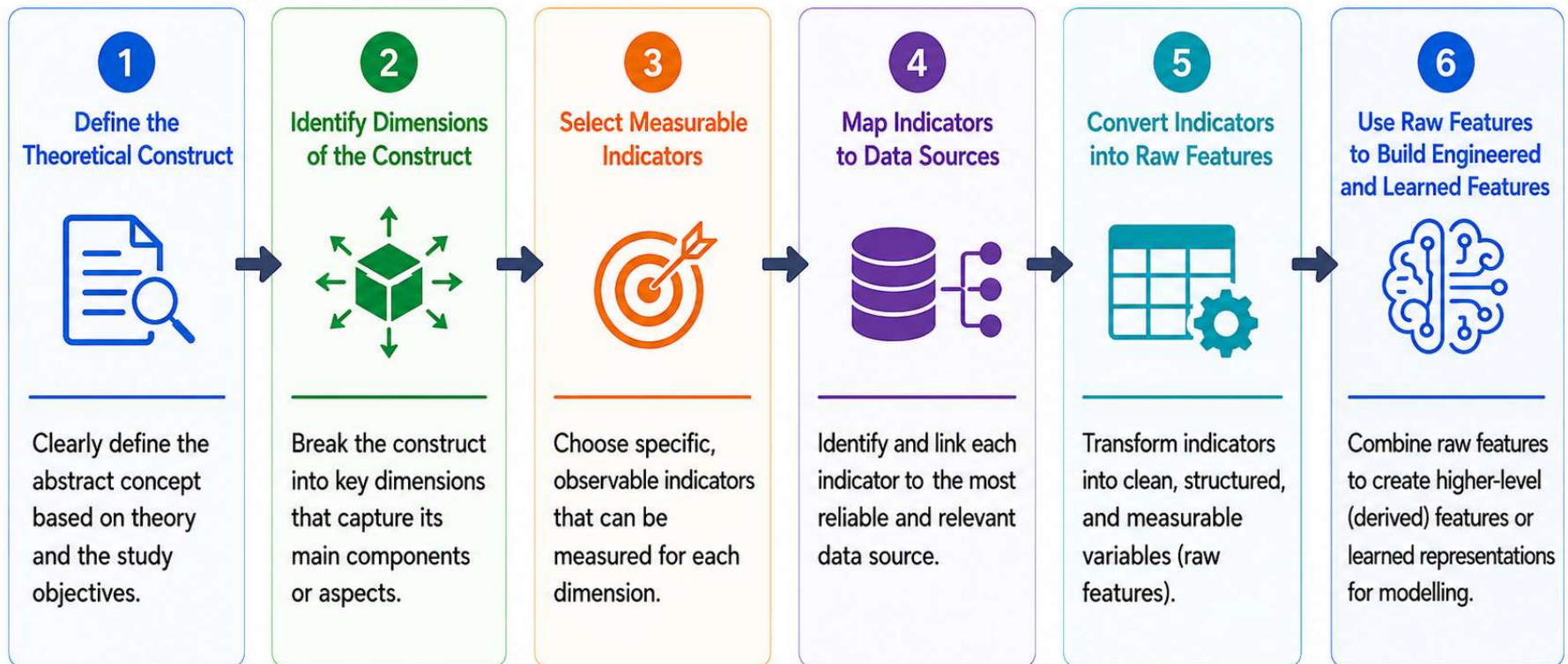
Criterion	What it Checks	Application to This Study
<i>Construct coverage</i>	Whether all key study variables are measured	AI adoption, DT exposure, employment impact, productivity, skills, readiness
<i>Reliability requirement</i>	Whether each construct has enough items for reliable measurement	Use about 3–5 questions per construct
<i>Research-objective alignment</i>	Whether each question supports a research objective or model variable	Remove questions that do not support forecasting, explanation, or validation
<i>Respondent burden</i>	Whether the survey is short enough to avoid fatigue	Banking employees should complete it in 10–15 minutes
<i>Sample-size suitability</i>	Whether the number of questions is manageable for n = 200 respondents	Avoid too many questions that weaken response quality or analysis

Suggested Question Allocation

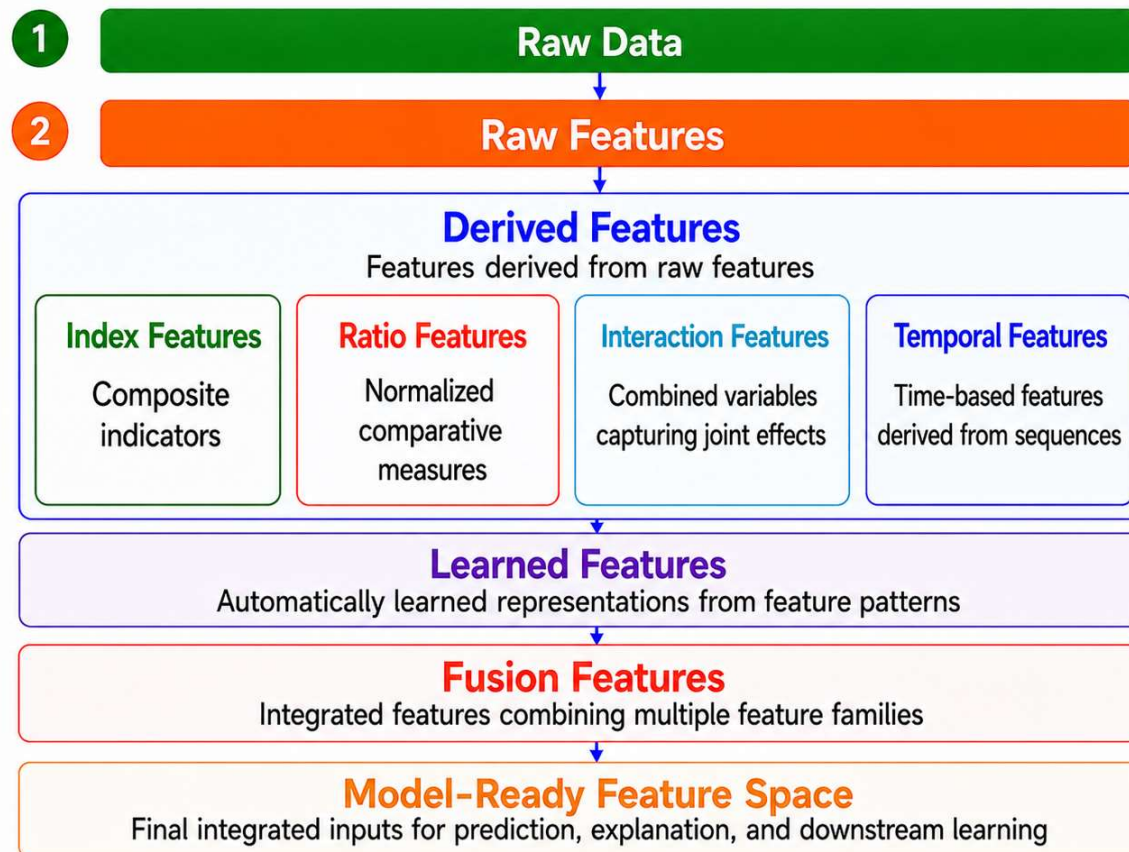
Section	Recommended Questions
Demographic and employment profile	5
Digital transformation exposure	5
AI adoption and usage	5
Employment and job impact	6
Workforce skills and reskilling	6
Productivity perception	5
Readiness, attitudes, and open comments	4
Total	35 questions

Recommended Operationalisation Process

Six Steps: From Theoretical Constructs to Model-Ready Features

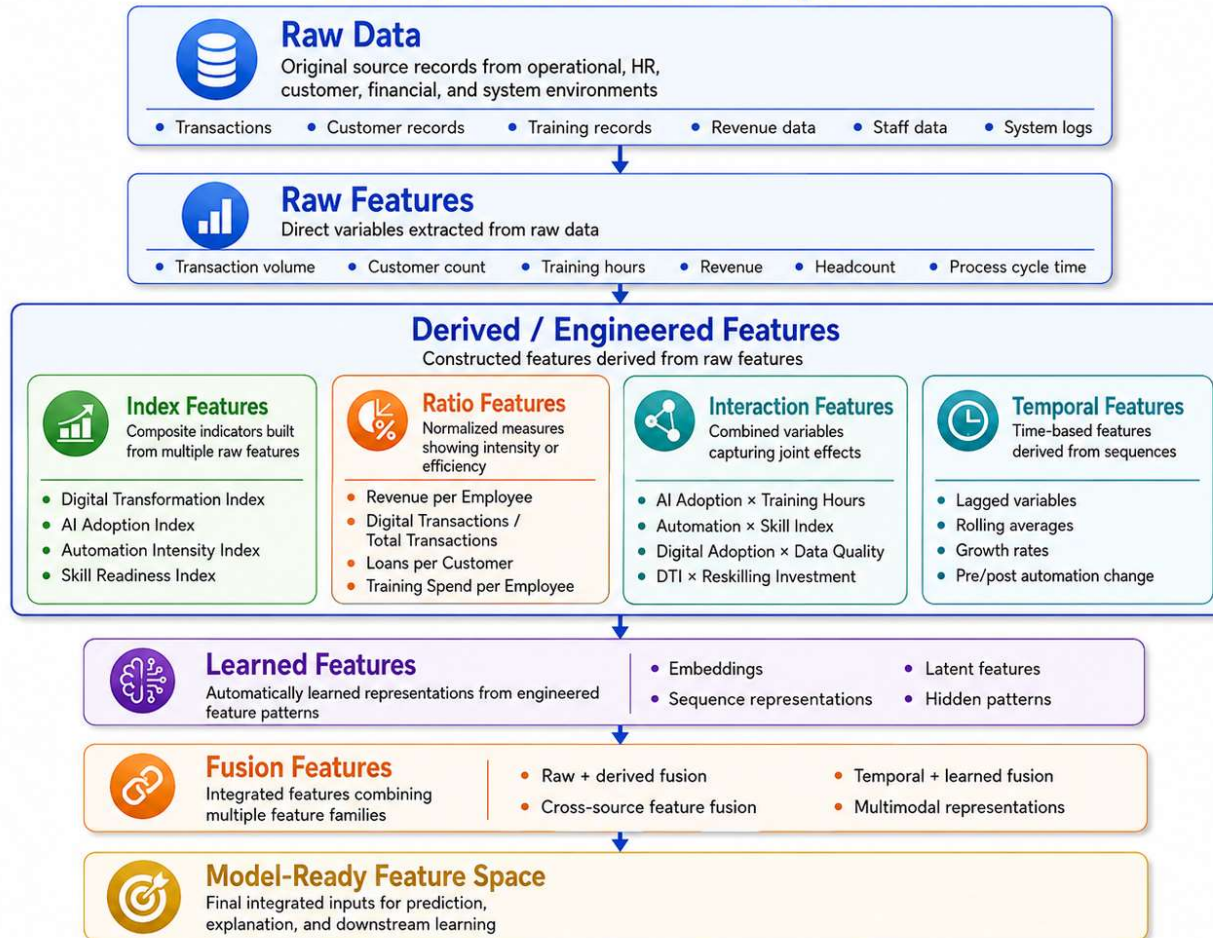


Feature Engineering Hierarchy



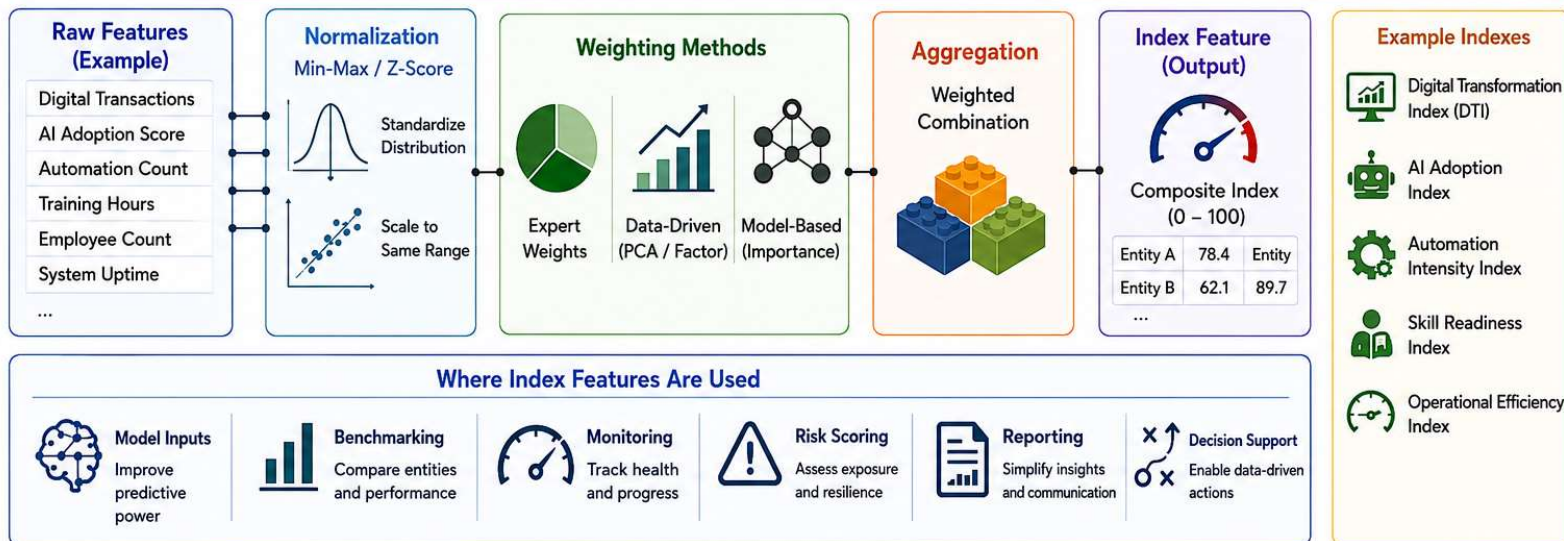
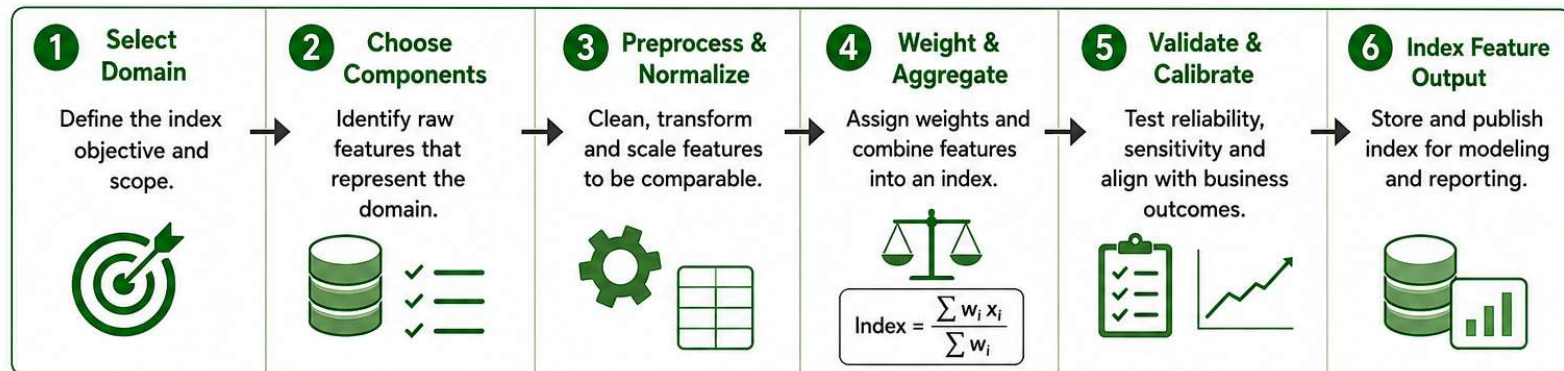
Feature Engineering Hierarchy

From raw data to fused model-ready features



Index Features: Methodology & Architecture

Build composite indicators that summarize multiple raw features into meaningful indices



Where Index Features Are Used



Model Inputs
Improve predictive power



Benchmarking
Compare entities and performance



Monitoring
Track health and progress



Risk Scoring
Assess exposure and resilience



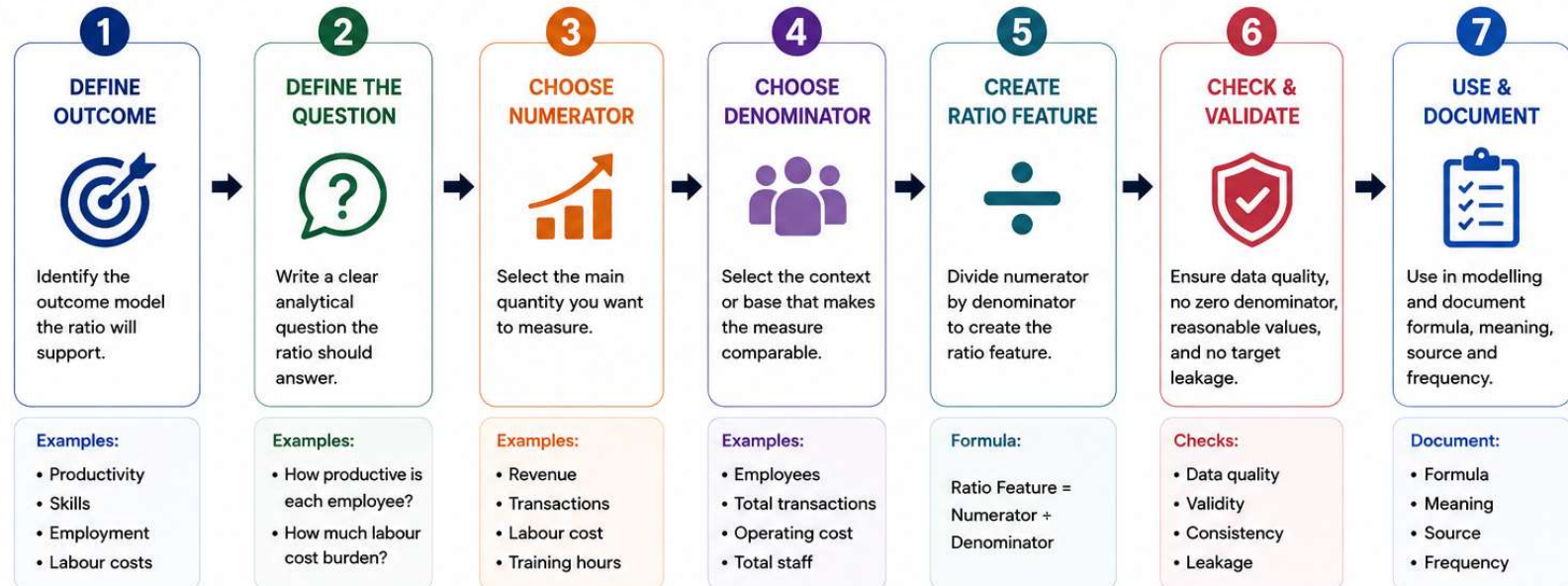
Reporting
Simplify insights and communication



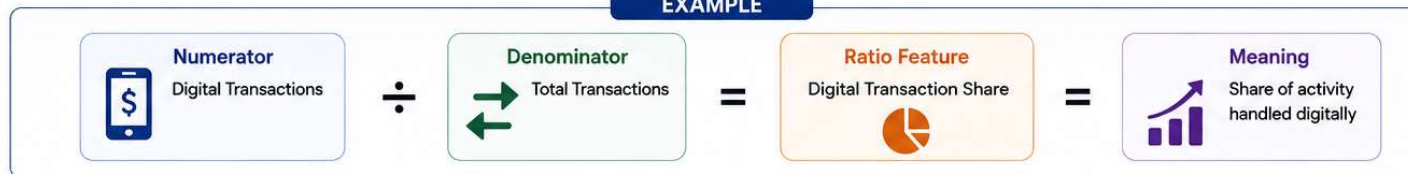
Decision Support
Enable data-driven actions

RATIO FEATURES: DESIGN METHODOLOGY (SIMPLIFIED)

Convert raw variables into relative, comparable and meaningful measures



EXAMPLE

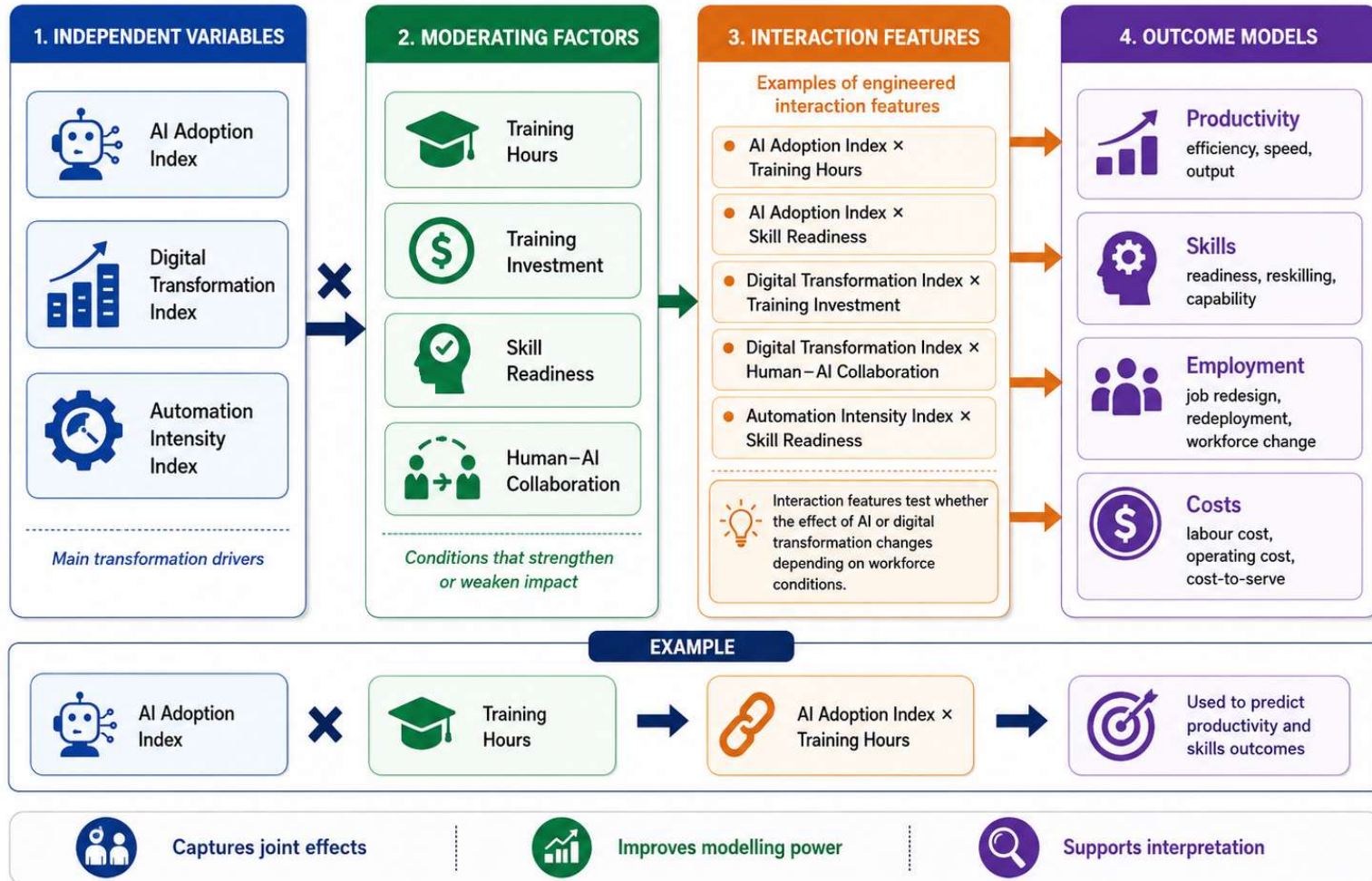


KEY TAKEAWAY:

Ratio features turn raw numbers into meaningful, comparable indicators that improve model performance, interpretation and decision-making.

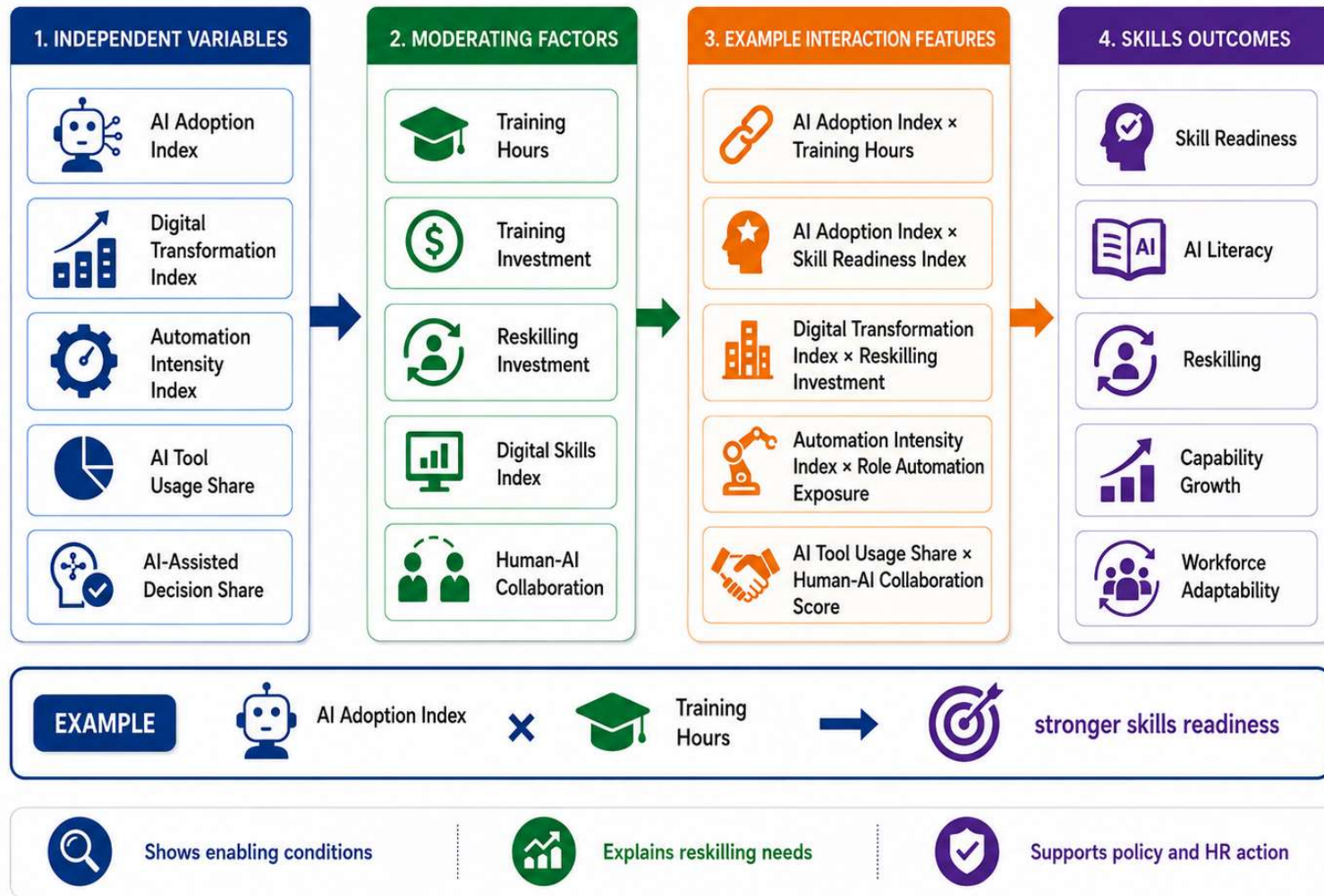
HOW INTERACTION FEATURES ARE ENGINEERED

Independent Variable × Moderating Factor → Interaction Feature → Outcome Model



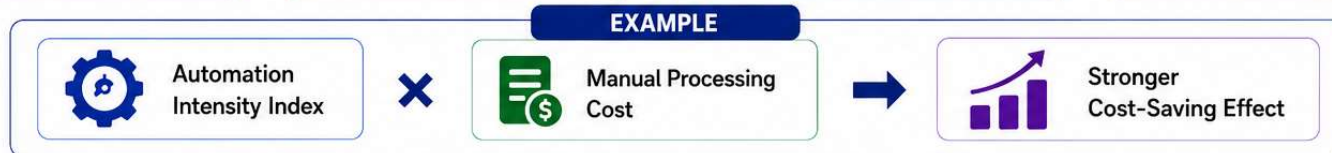
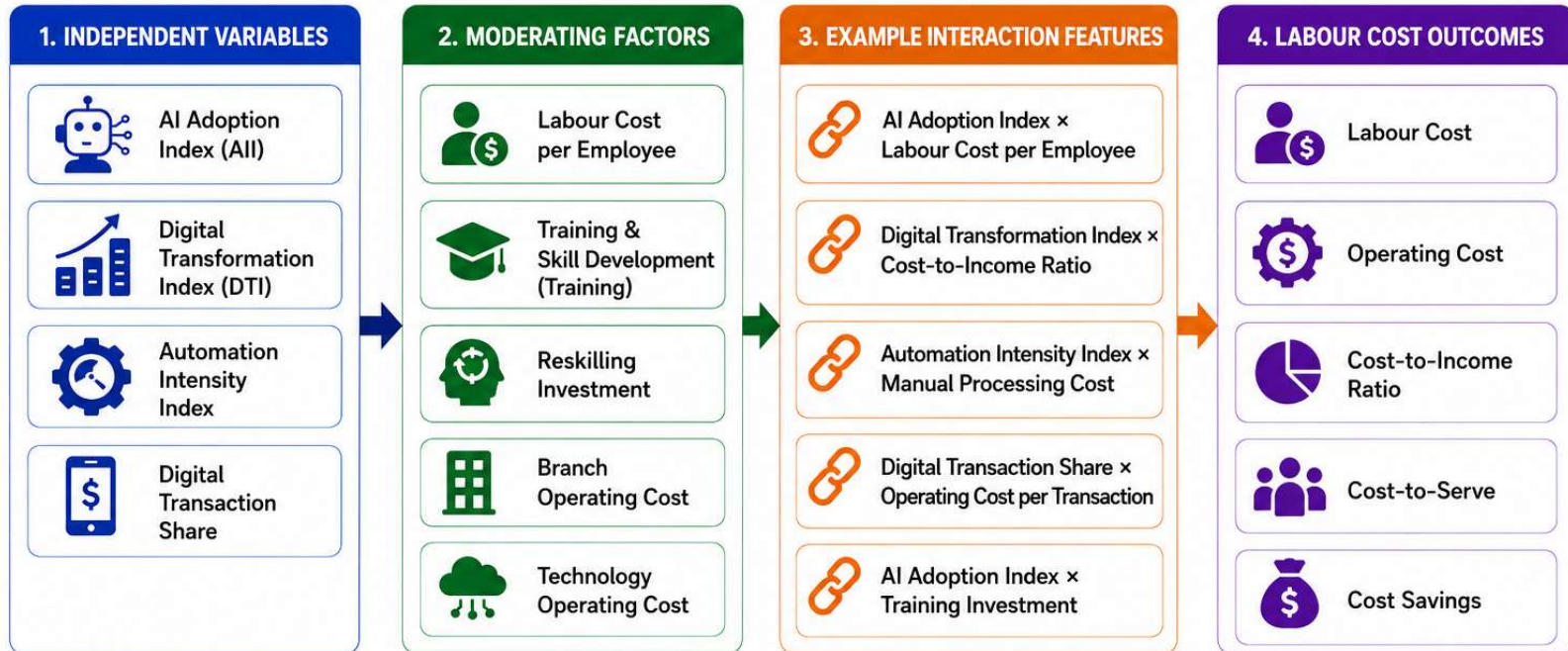
SKILLS INTERACTION FEATURES

How AI and digital transformation features are engineered for workforce skills outcomes



LABOUR COSTS INTERACTION FEATURES

How AI and digital transformation features are engineered for labour cost and operating cost outcomes



Reveals cost drivers

Identifies which factors most influence labour costs.



Shows efficiency gains

Highlights interaction effects that drive cost efficiencies.

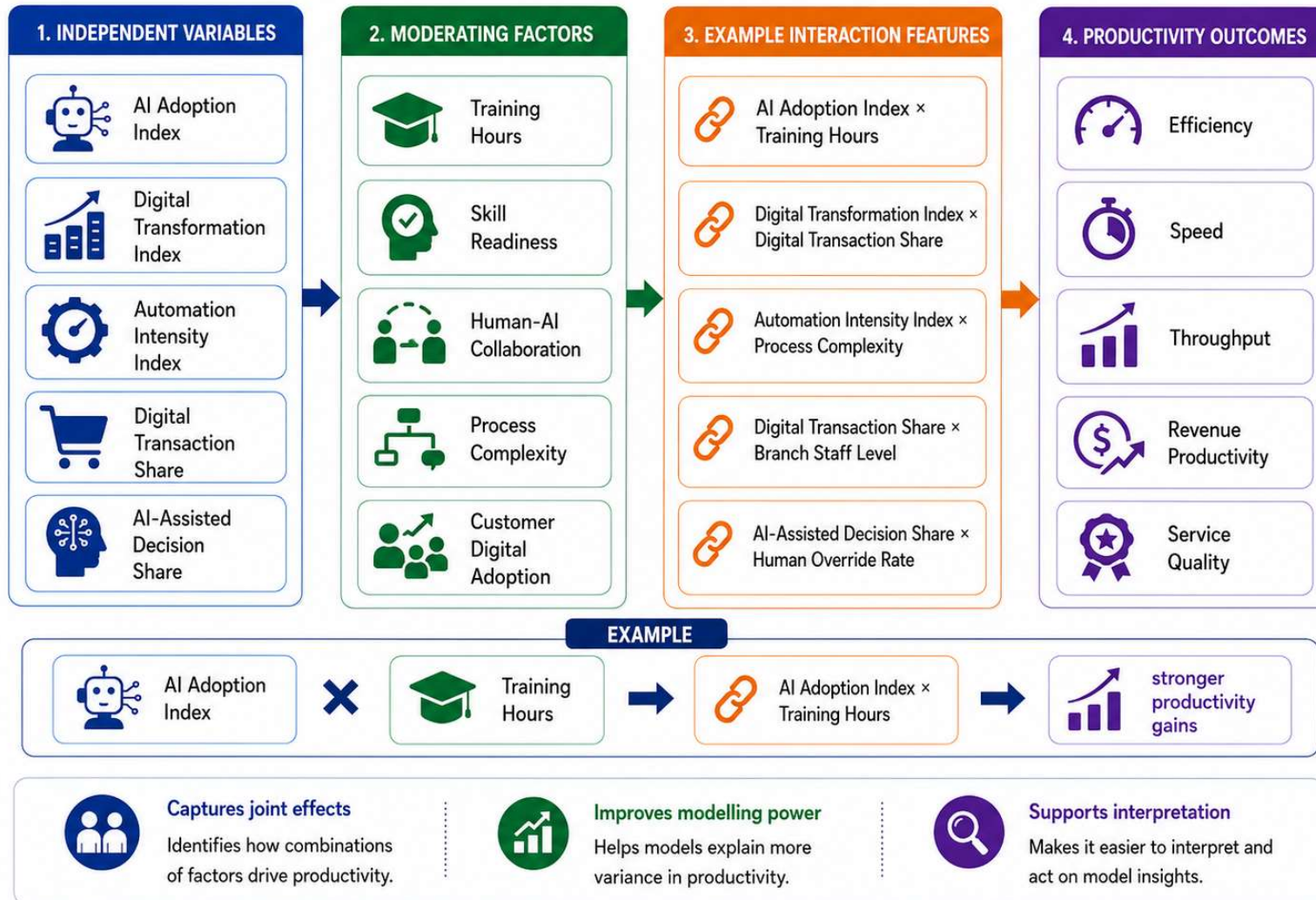


Supports financial planning

Provides insights to plan budgets and improve profitability.

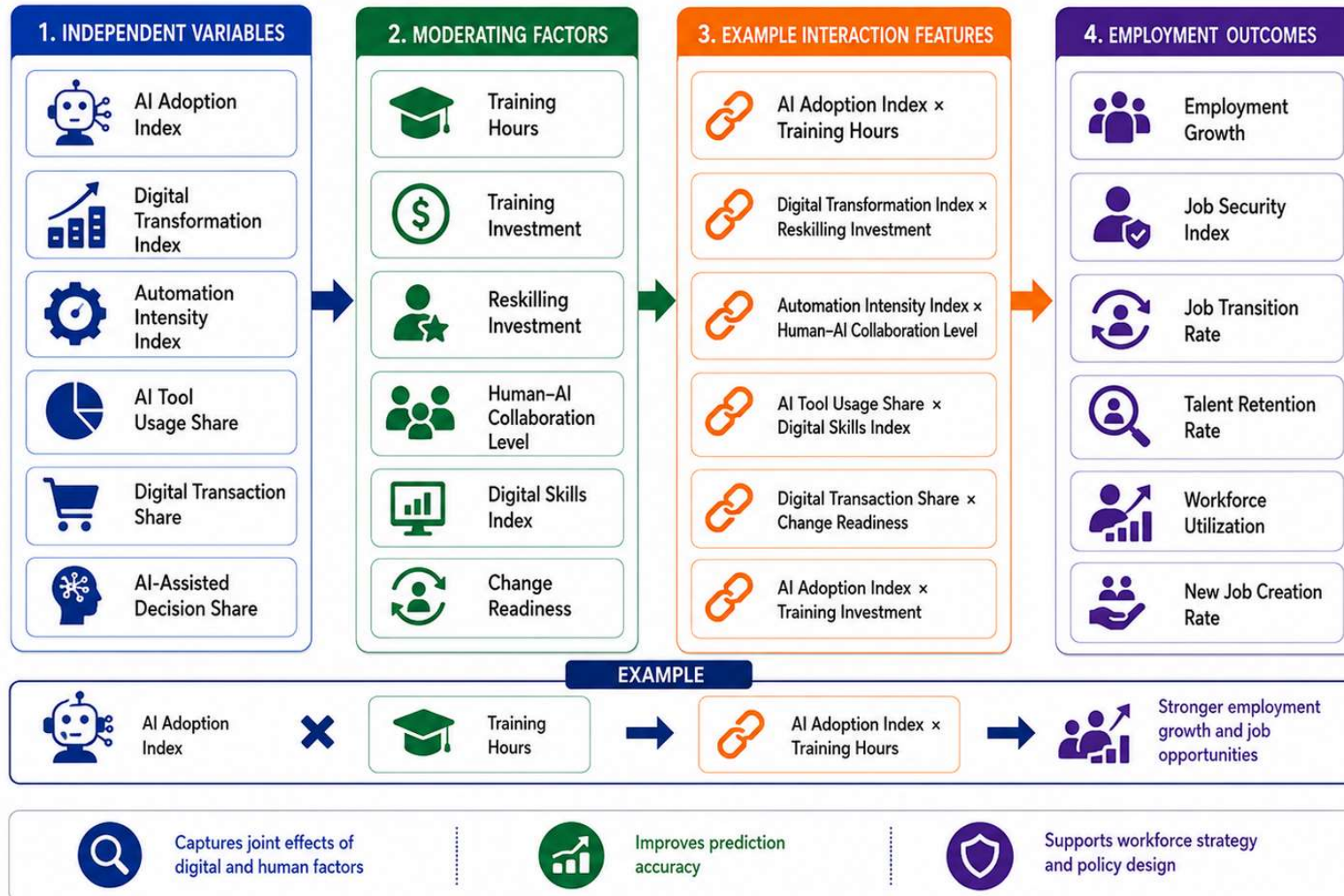
PRODUCTIVITY INTERACTION FEATURES

How AI and digital transformation features are engineered for productivity outcomes



EMPLOYMENT INTERACTION FEATURES

How AI and digital transformation features are engineered for employment outcomes



Temporal Features Methodology & Architecture

How time-based features are designed, processed, and used in predictive systems



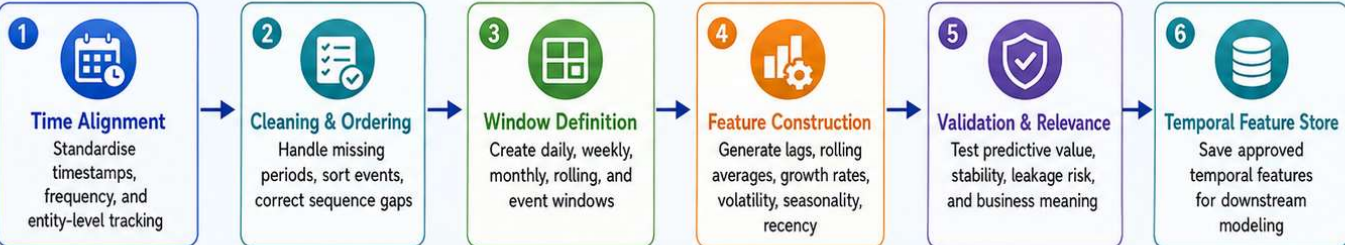
Temporal Data Foundation

Time-stamped observations collected across periods, events, or sequences

- Transactions by date
- Customer activity history
- Training records over time
- Revenue by month
- Staff movement
- System event logs

Methodology for Temporal Feature Engineering

Step-by-step process for constructing time-aware features



Common Temporal Features

- Lagged variables
- Moving averages
- Growth/change rates
- Cumulative totals
- Trend indicators
- Seasonality markers
- Recency/frequency
- Pre/post event change

Temporal Feature Architecture

Operational architecture for using time-based features

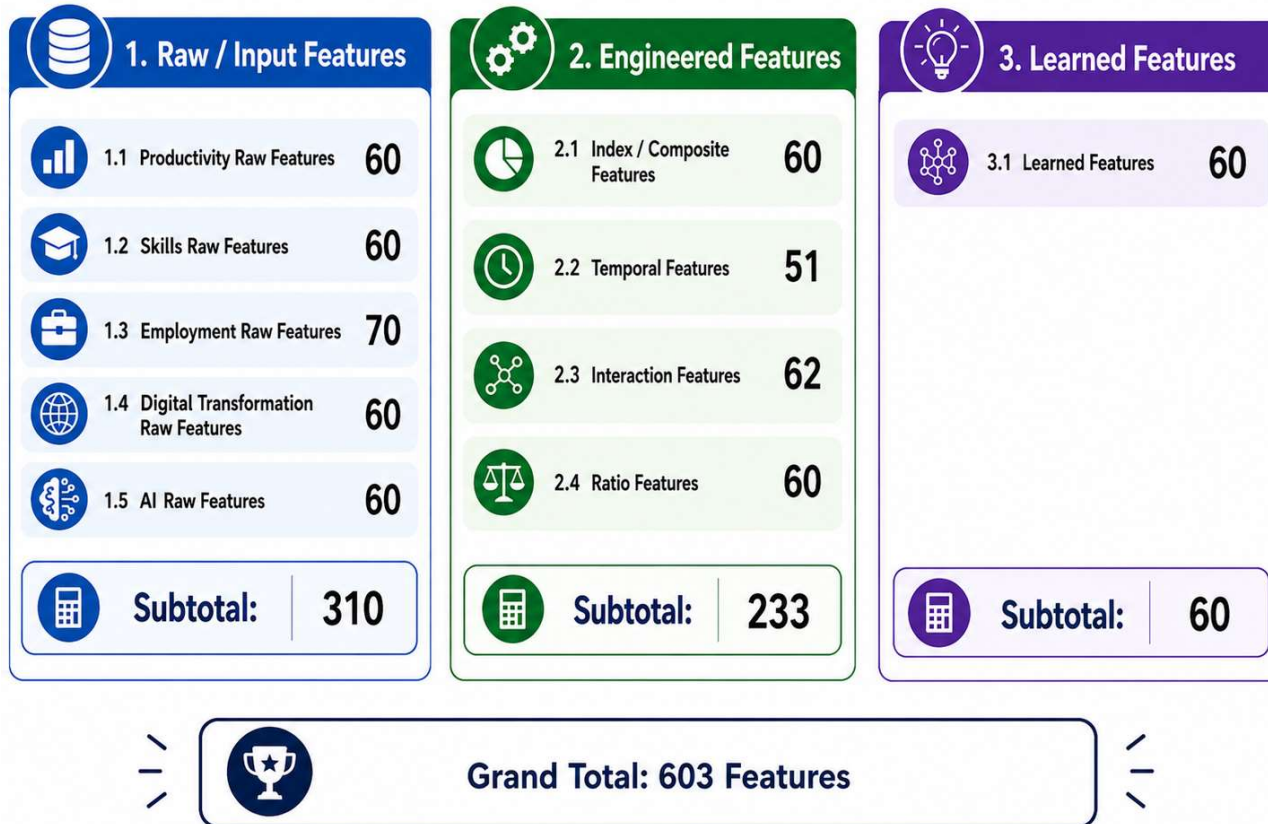


Outcome

Model-ready temporal features that preserve order, change, memory, and time-based behaviour

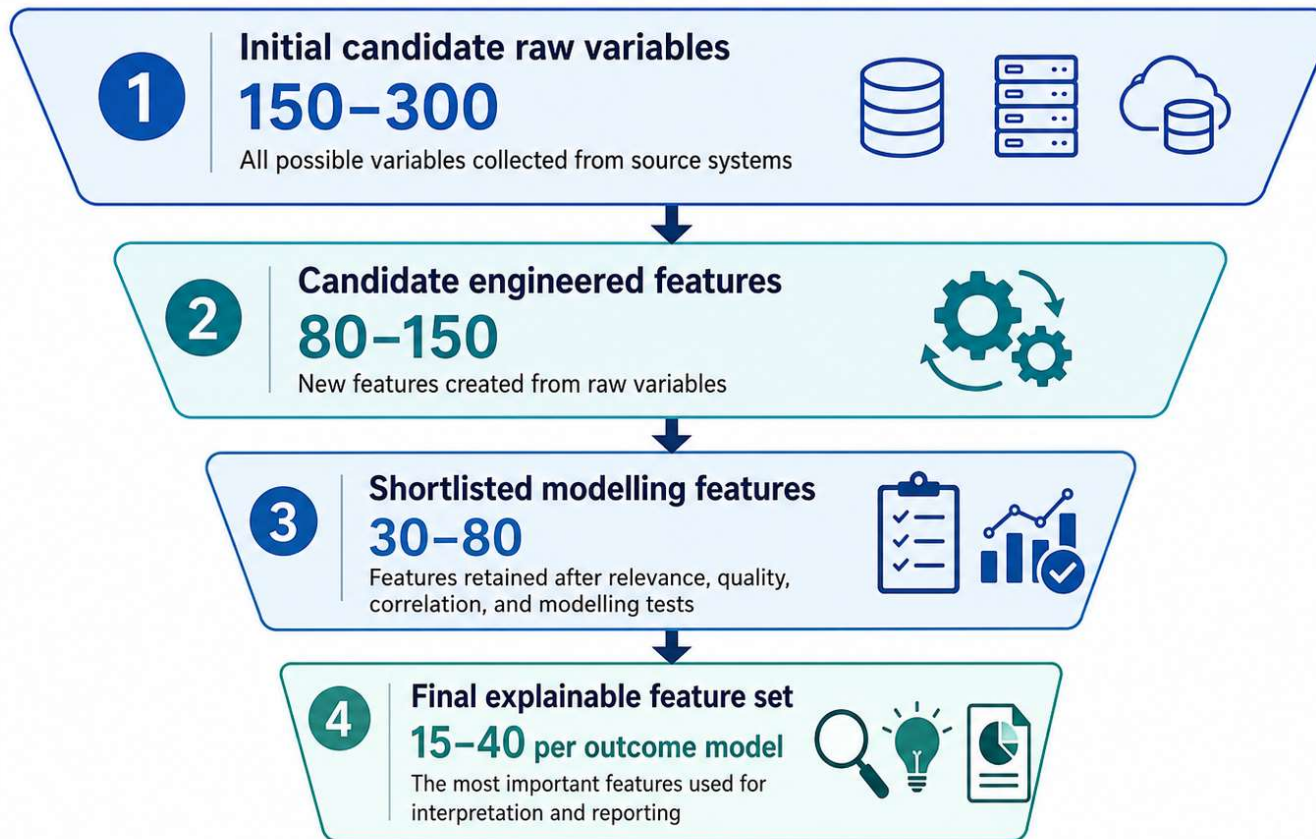
Feature Categories and Quantities

Updated Feature Inventory for the AI and Digital Transformation Research Framework



Feature Reduction Journey

From broad raw variables to the final explainable feature set



Start broad → engineer intelligently → shortlist carefully → explain clearly

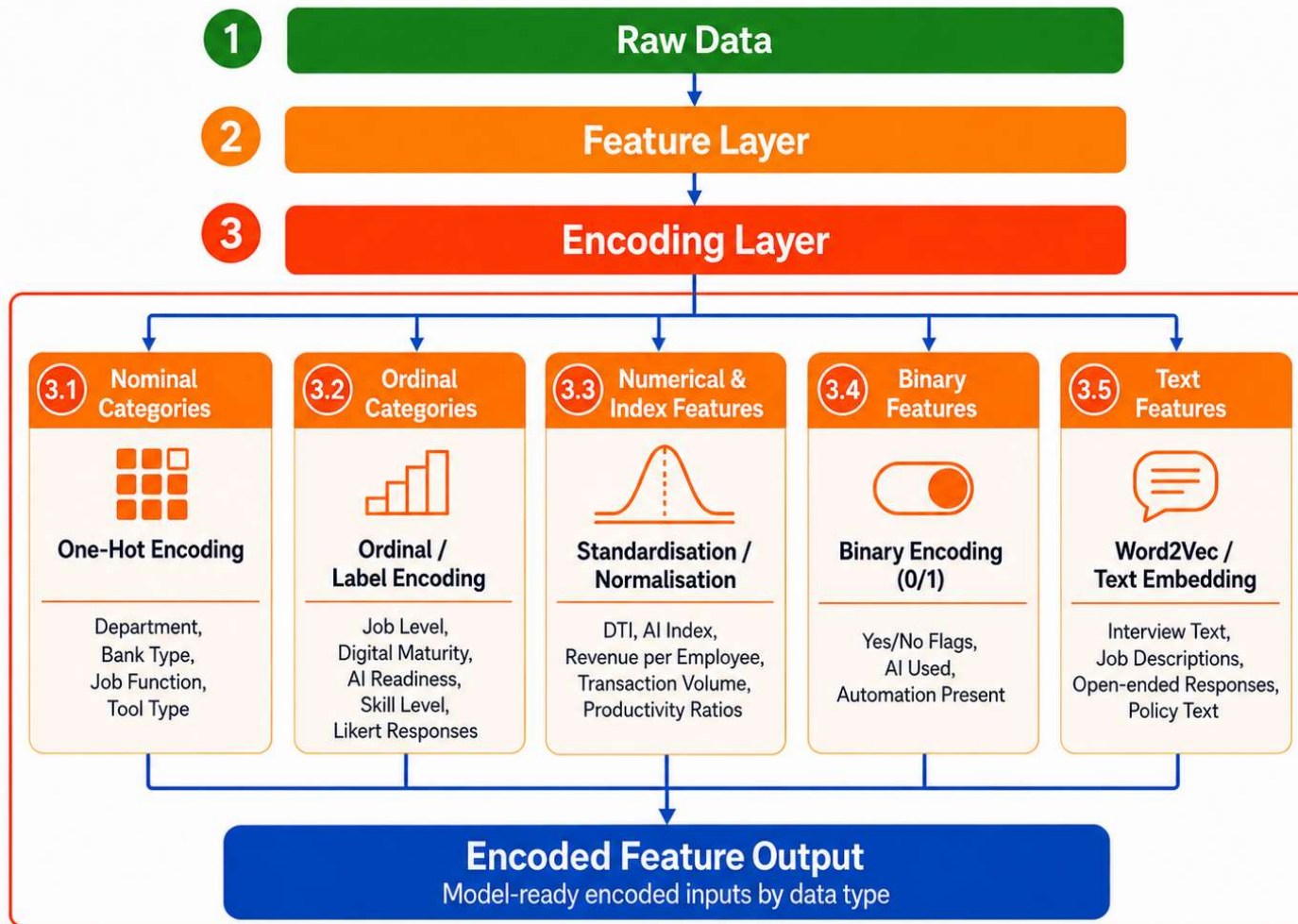
Outcome Models and Final Feature Sets

Each outcome model may have its own final explainable feature set.

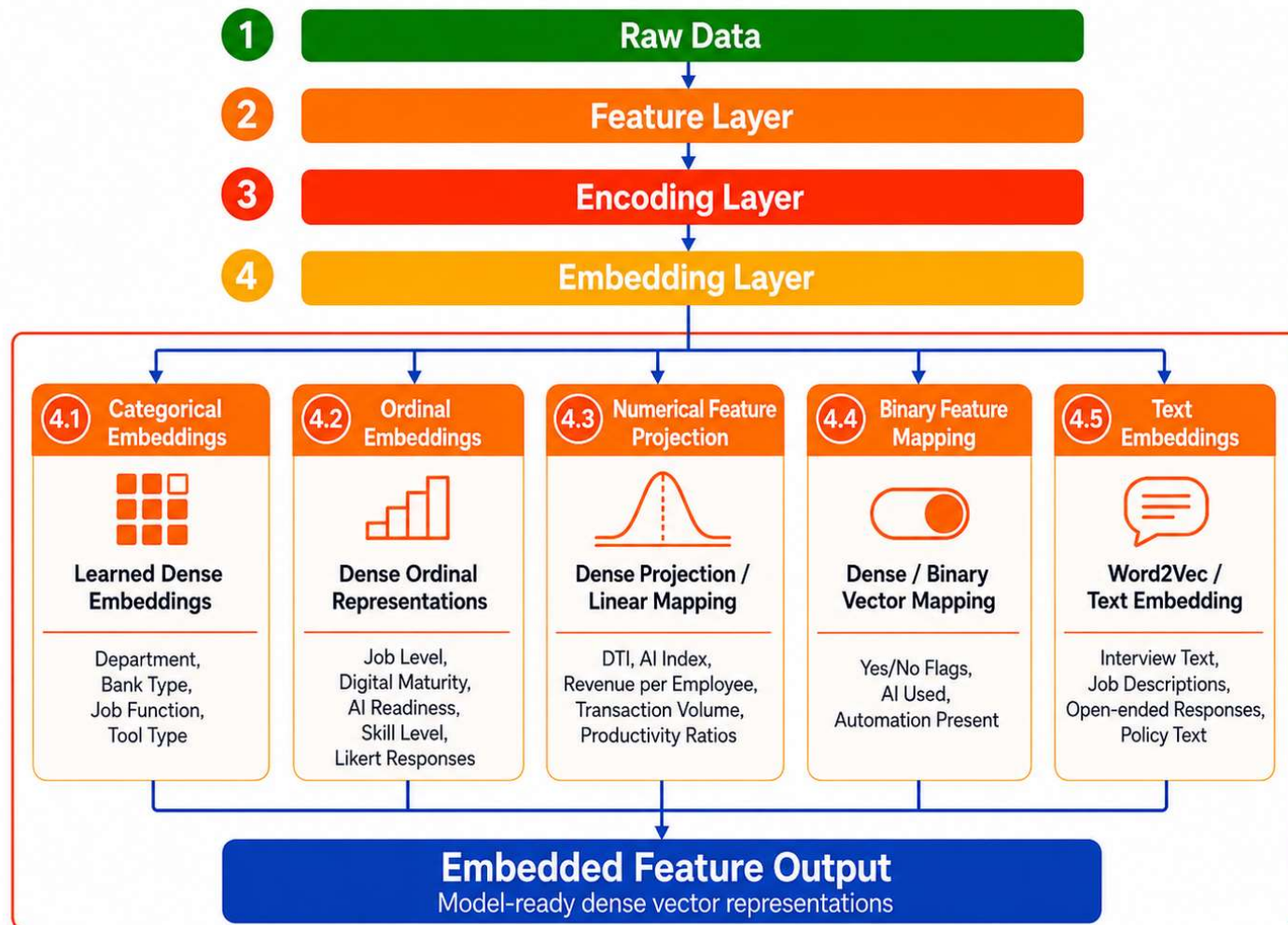
Outcome model	Final feature set
 Productivity model	15–40 productivity-relevant features
 Skills model	15–40 skills-relevant features
 Employment model	15–40 employment-relevant features

 *Different models can follow different feature reduction journeys and final feature sets.*

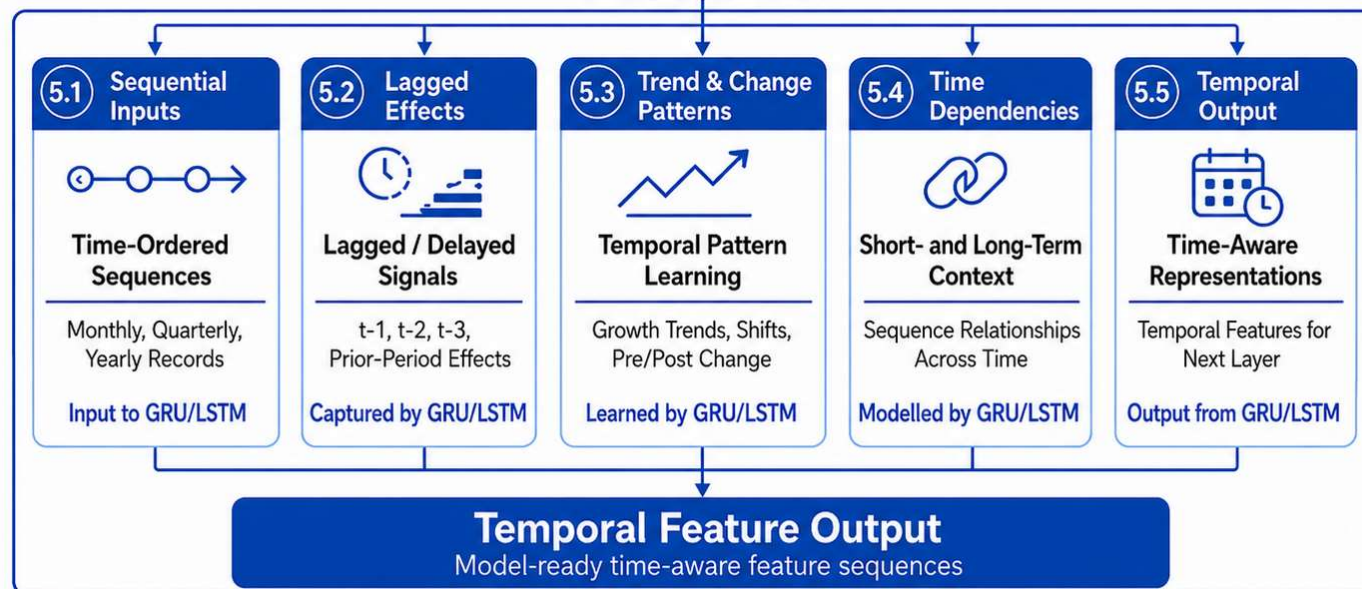
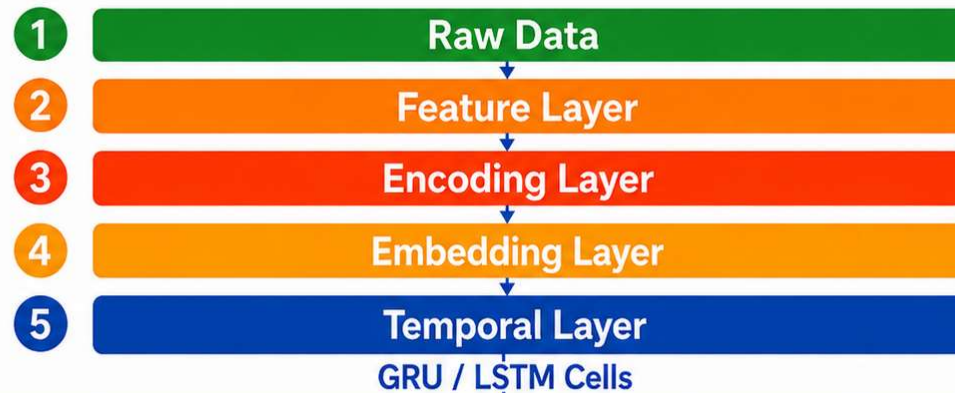
Encoding Layer Architecture



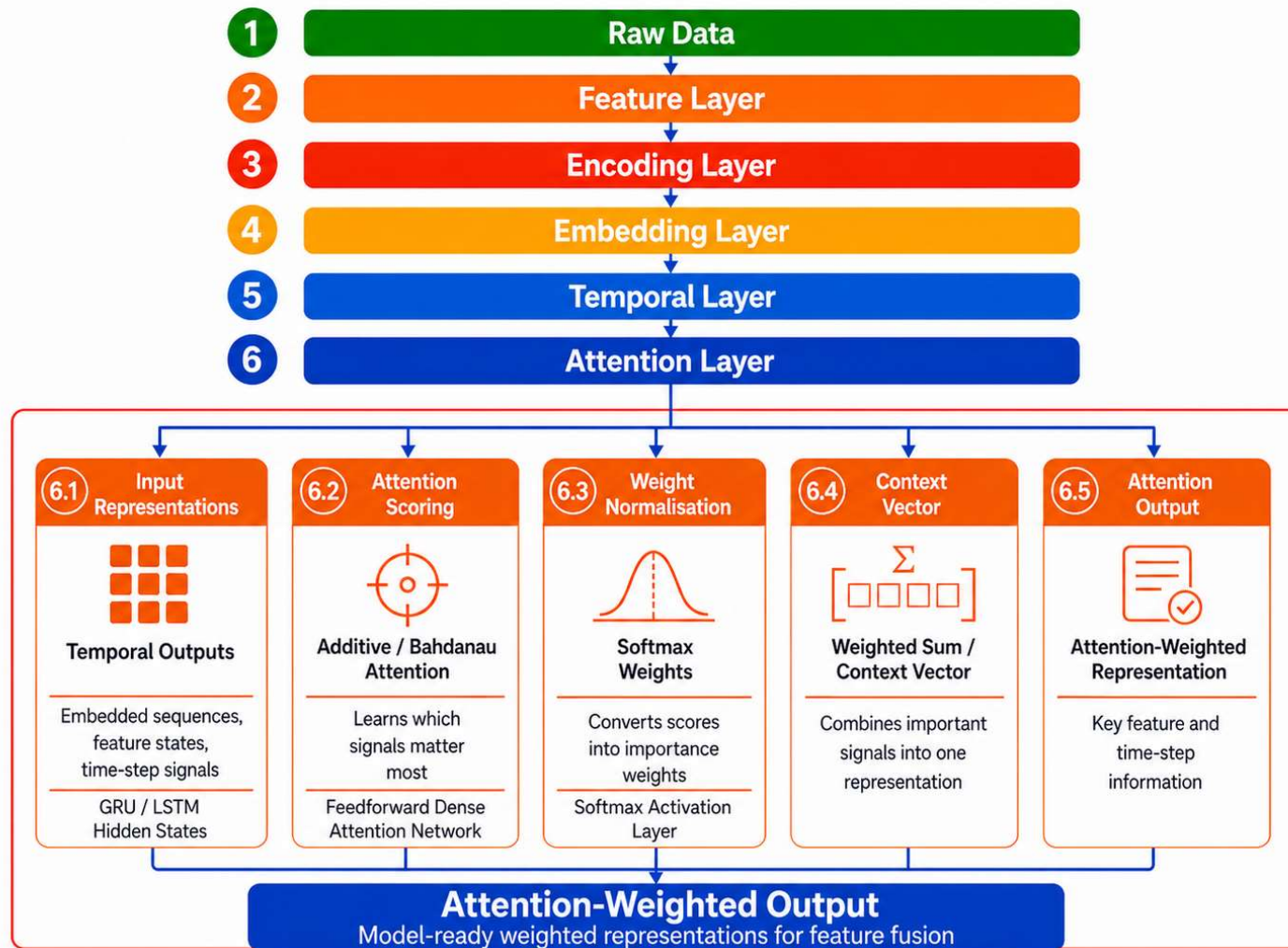
Embedding Layer Architecture



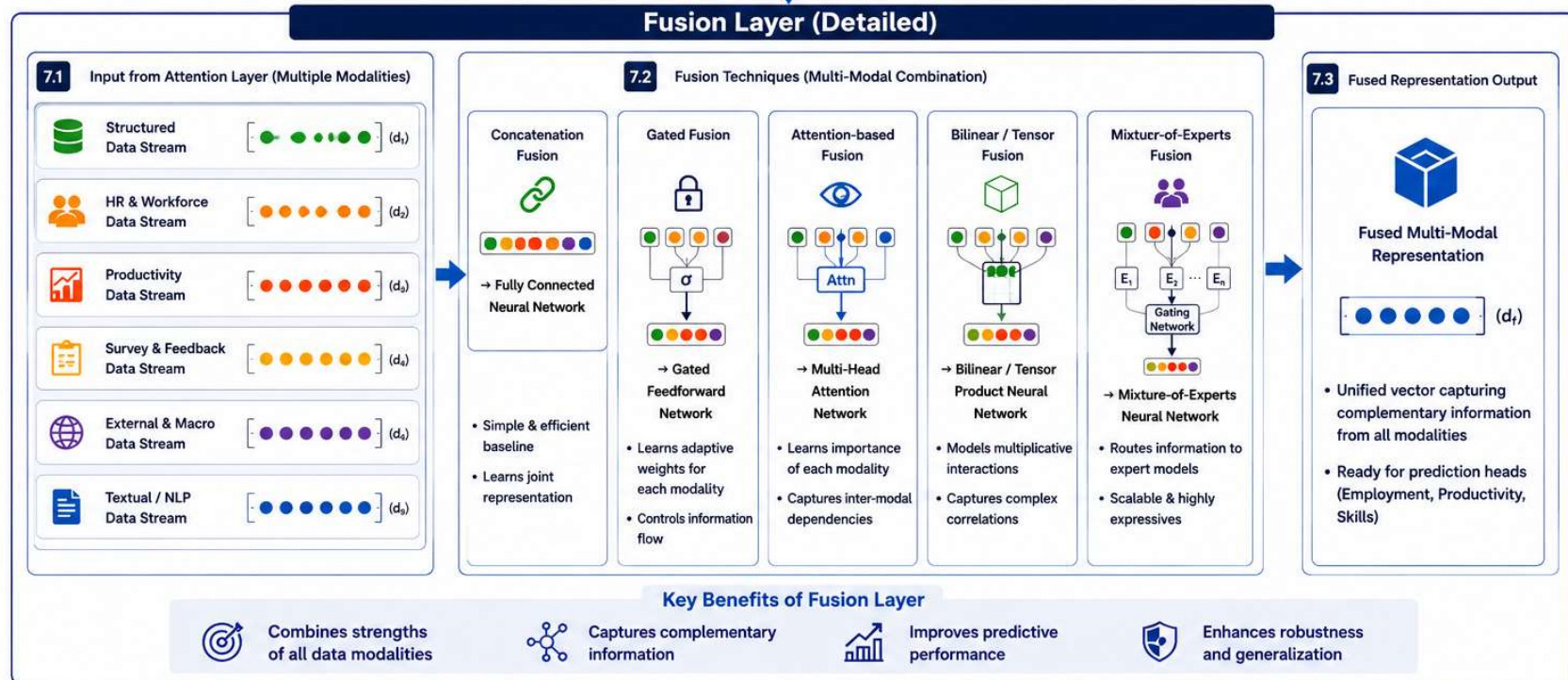
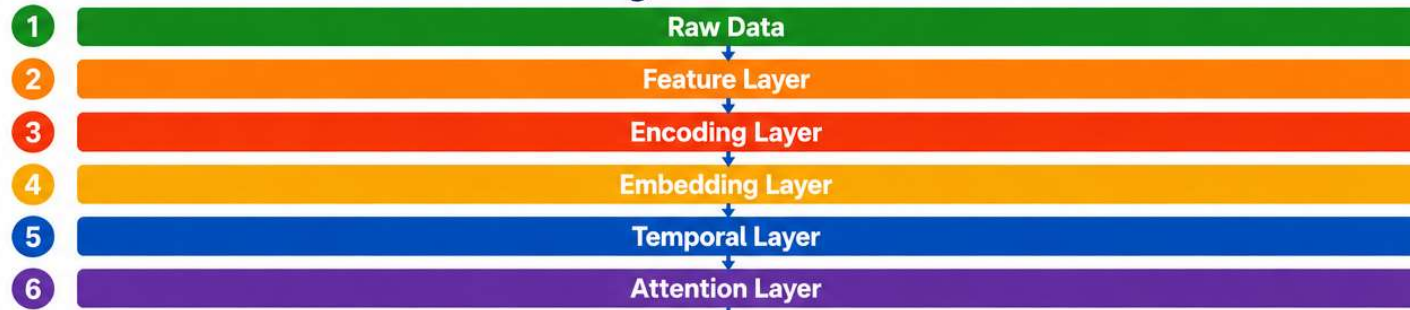
Temporal Layer Architecture



Attention Layer Architecture



Fusion Layer Architecture



Fusion Layer Architecture

6

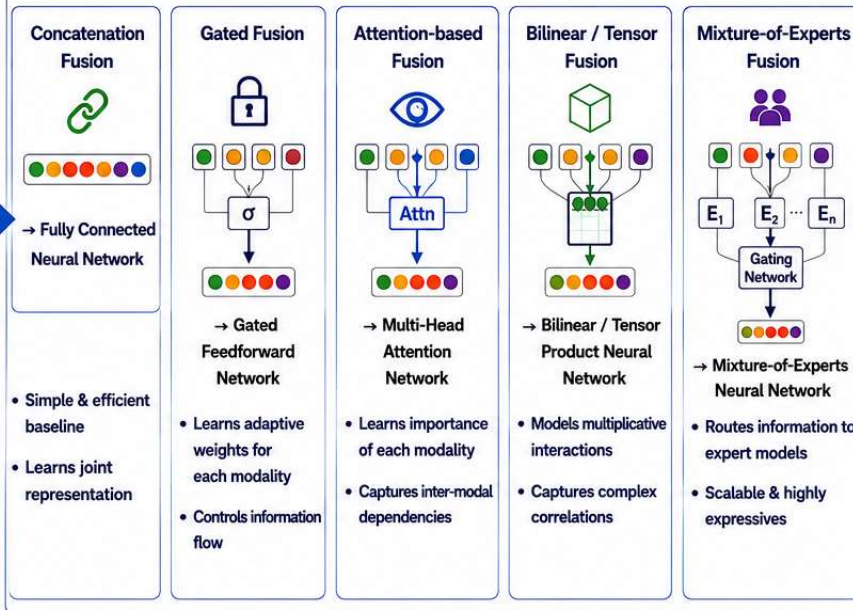
Attention Layer

Fusion Layer (Detailed)

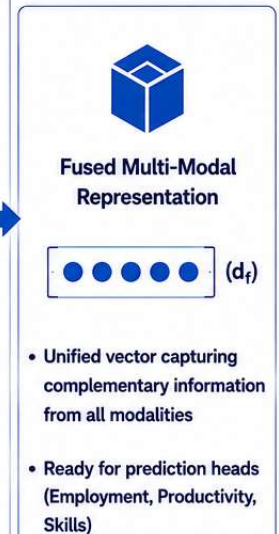
7.1 Input from Attention Layer (Multiple Modalities)



7.2 Fusion Techniques (Multi-Modal Combination)



7.3 Fused Representation Output



Key Benefits of Fusion Layer



Combines strengths of all data modalities



Captures complementary information



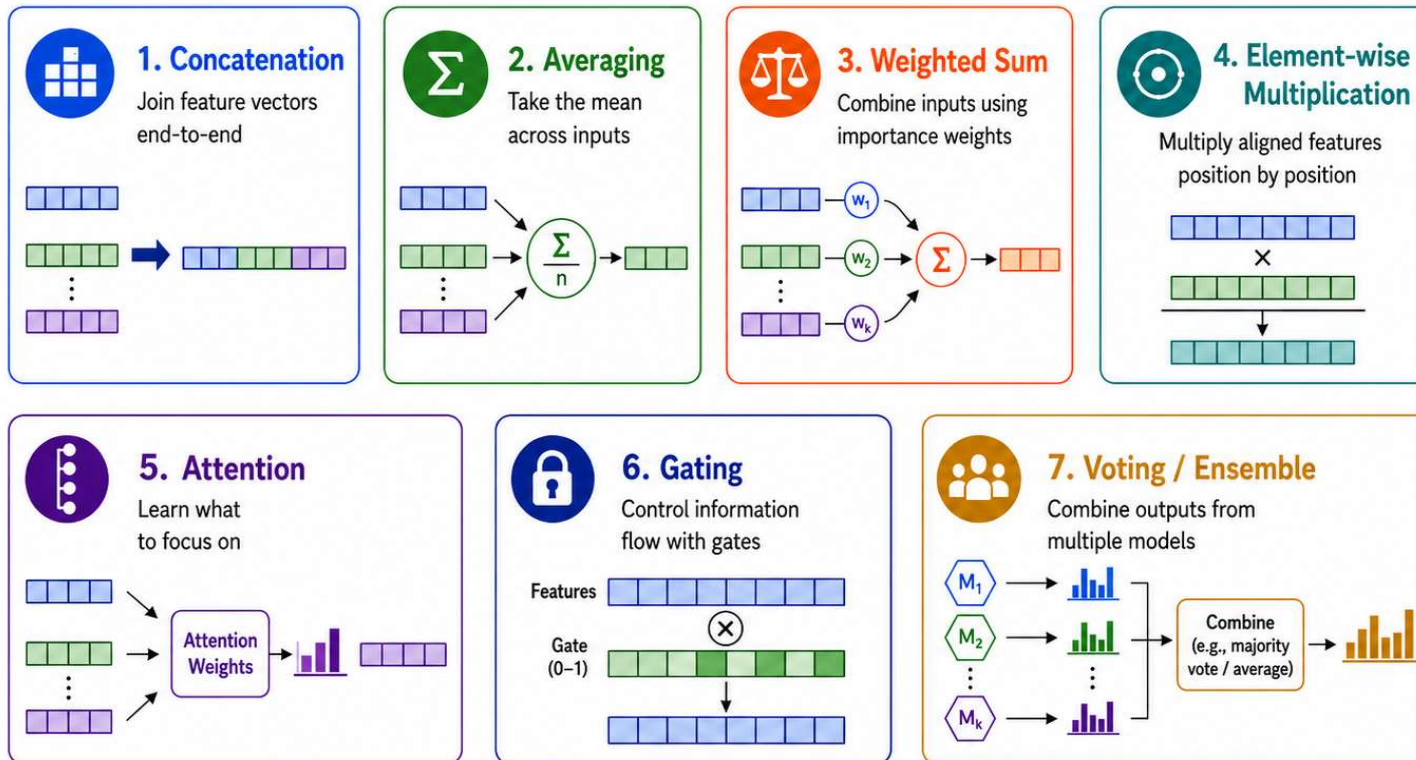
Improves predictive performance



Enhances robustness and generalization

Common Fusion Operations

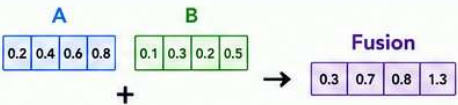
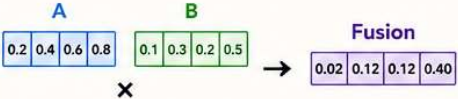
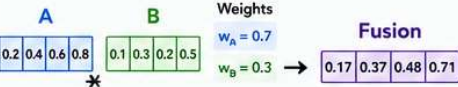
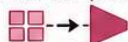
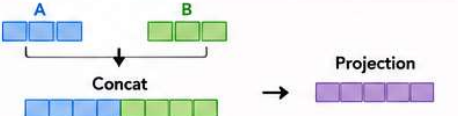
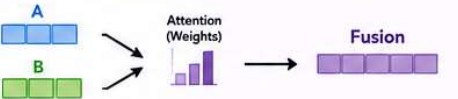
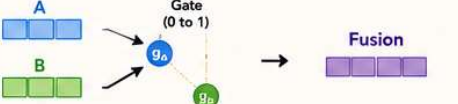
Core methods for combining multiple feature sets or model outputs



These operations can be used alone or combined in more advanced fusion strategies.

Common Feature Fusion Operations in AI

Ways to combine features and the AI models/technologies that implement them

Fusion Operation	How It Works (In Simple Terms)	Visual Example	AI Models / Technologies	Best Used For
1. Concatenation (Feature Stacking) 	Stack features from different sources into a single longer vector.		 - Fully Connected (Dense) Layers - MLP (Multi-Layer Perceptron) - Deep Neural Networks (DNN)	General purpose, simple and effective baseline fusion.
2. Element-wise Sum (Addition) 	Add features of the same dimension element by element.		 - Residual Connections (ResNet) - Gated Recurrent Units (GRU) - Transformers (Add & Norm)	Same-dimension features, residual learning, stability.
3. Element-wise Multiplication (Product) 	Multiply features of the same dimension element by element.		 - Gated Mechanisms (LSTM/GRU) - Bilinear Models - Feature Interaction Layers	Capture interactions, feature gating, importance modeling.
4. Weighted Sum (Linear Combination) 	Combine features using learnable weights before summing.		 - Attention Mechanisms - Mixture of Experts (MoE) - Learnable Weighted Layers	When sources have different importance; adaptive weighting.
5. Concatenation + Projection (Learned Compression) 	Concatenate features then project to a lower dimension using a learned layer.		 - Fully Connected + Linear Layer - Autoencoders - Transformers (Input Projection)	High-dimensional fusion, reduce redundancy, improve efficiency.
6. Attention-based Fusion (Contextual Weighting) 	Use attention to learn how much each source should contribute in context.		 - Transformers (Multi-Head Attention) - Cross-Attention - Self-Attention Mechanisms	Sequences, text, images, multimodal data; focus on relevant information.
7. Gating-based Fusion (Learned Gate) 	Learn a gate (0-1) to control how much information from each source is passed.		 - Gated Fusion Networks - LSTM/GRU Gates - Highway Networks	Handle noisy/missing sources; adaptive information flow.

 **Key Takeaway:** Choose the fusion operation based on data characteristics, dimensionality, and the relationships you want the model to learn.

Choosing the Right Fusion Operation

Selection criteria for picking the best way to combine features or model outputs

1. Key Selection Criteria



Dimension Match

Are inputs same size or different sizes?



Source Importance

Are all sources equally important?



Interaction Need

Do you need cross-feature interactions?



Context Dependence

Should importance change by context?



Noise / Missing Data

Are some sources unreliable or absent?



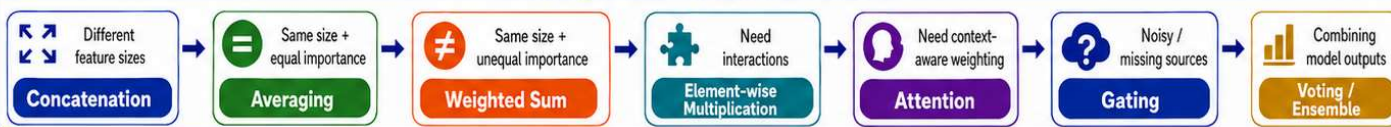
Fusion Level

Are you combining features or model outputs?

2. Operation Selection Guide

 Concatenation	Best when: different feature groups, simple baseline, preserve all information	Technology: Dense layers, MLP, DNN
 Averaging	Best when: same-dimension inputs, equal importance, simple smoothing	Technology: Ensemble averaging, mean pooling
 Weighted Sum	Best when: sources have unequal importance, weights known or learnable	Technology: Learnable linear layers, attention weights
 Element-wise Multiplication	Best when: same-dimension features, interaction or gating effects matter	Technology: Bilinear layers, feature interaction layers, gates
 Attention	Best when: importance should change by context, sequences or multimodal data	Technology: Transformers, self-attention, cross-attention
 Gating	Best when: need to suppress noisy or missing sources, adaptive information flow	Technology: LSTM/GRU gates, highway networks, gated fusion
 Voting / Ensemble	Best when: combining outputs from separate models, improve robustness	Technology: Bagging, boosting, majority vote, stacking

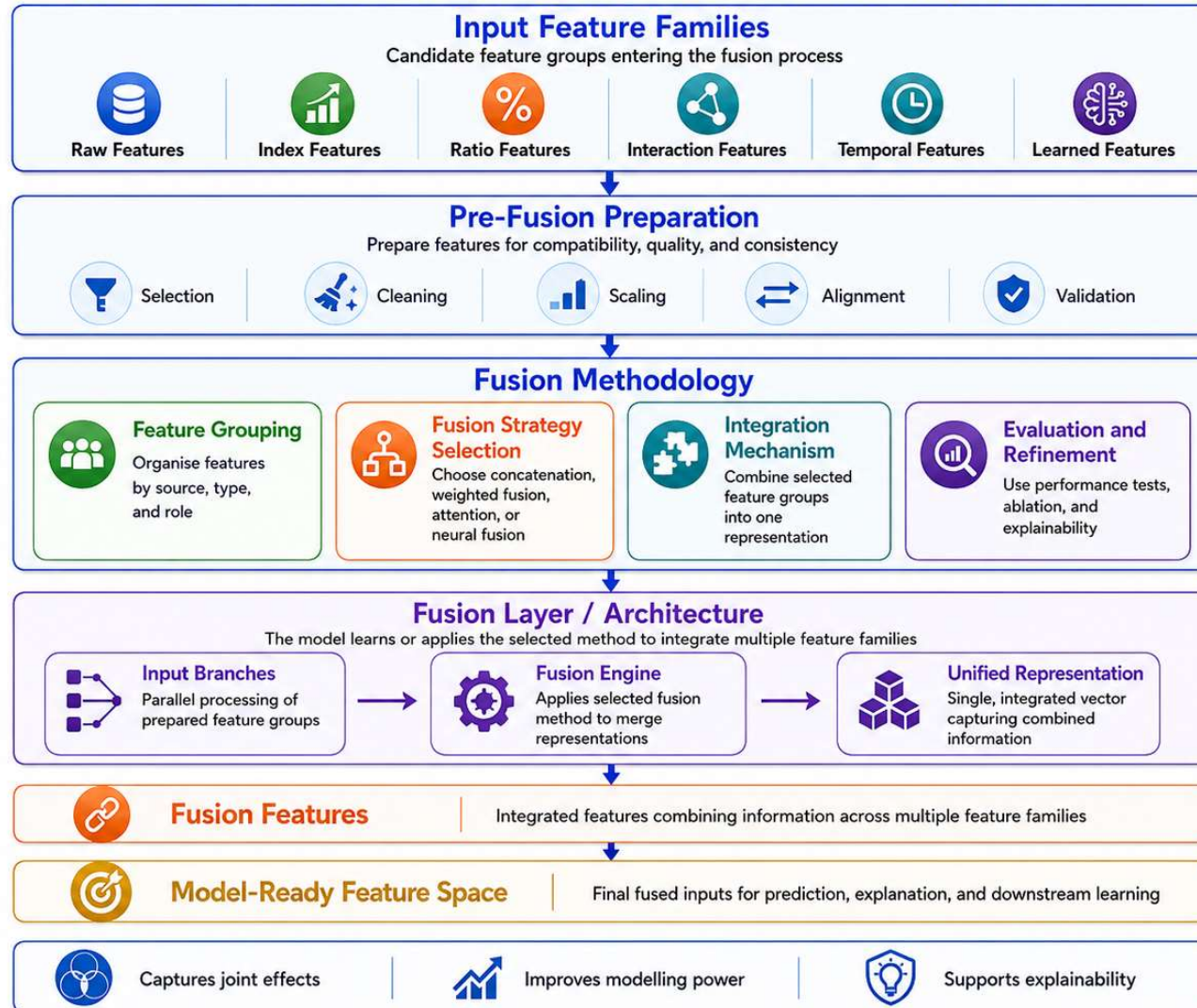
3. Quick Decision Rules



Choose the simplest operation that matches your data structure, source quality, and learning objective.

Fusion Features Methodology and Architecture

How multiple feature families are integrated into a unified model-ready representation



Fusion Features: Different Ways to Do Fusion

How multiple feature families are combined into unified model-ready representations

1. Fusion Inputs

Feature families available for integration



Raw Features



Index Features



Ratio Features



Interaction Features



Temporal Features



Learned Features

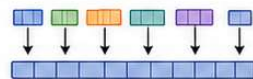
2. Different Ways to Do Fusion



1. Early Fusion

Combine features before modeling

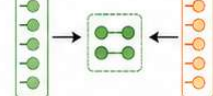
- concatenate inputs
- single joint model
- simple and common



2. Intermediate Fusion

Fuse hidden representations inside the model

- layer-level merging
- deep networks
- captures interactions



3. Late Fusion

Fuse outputs from separate models

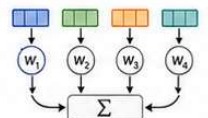
- model-level combination
- voting / averaging
- good for modular systems



4. Weighted Fusion

Assign importance to each source

- static or dynamic weights
- weighted sum
- balances source influence



5. Attention-Based Fusion

Learn what to focus on

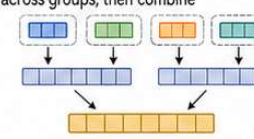
- context-aware weighting
- adaptive importance
- strong for complex patterns



6. Hierarchical / Hybrid Fusion

Fuse in stages across groups, then combine

- group-level fusion
- multi-stage integration
- handles many sources



3. Common Fusion Operations



Concatenation



Averaging



Weighted Sum



Element-wise Multiplication



Attention



Gating



Voting/
Ensemble

4. Fusion Architecture Flow



A. Input Feature Families

- raw
- temporal
- learned
- index



Alignment & Scaling

- normalization
- dimension alignment
- missing handling



C. Fusion Layer / Strategy

- early / late
- attention / weighted



D. Modeling Layer

- neural nets
- trees
- regression
- sequence models



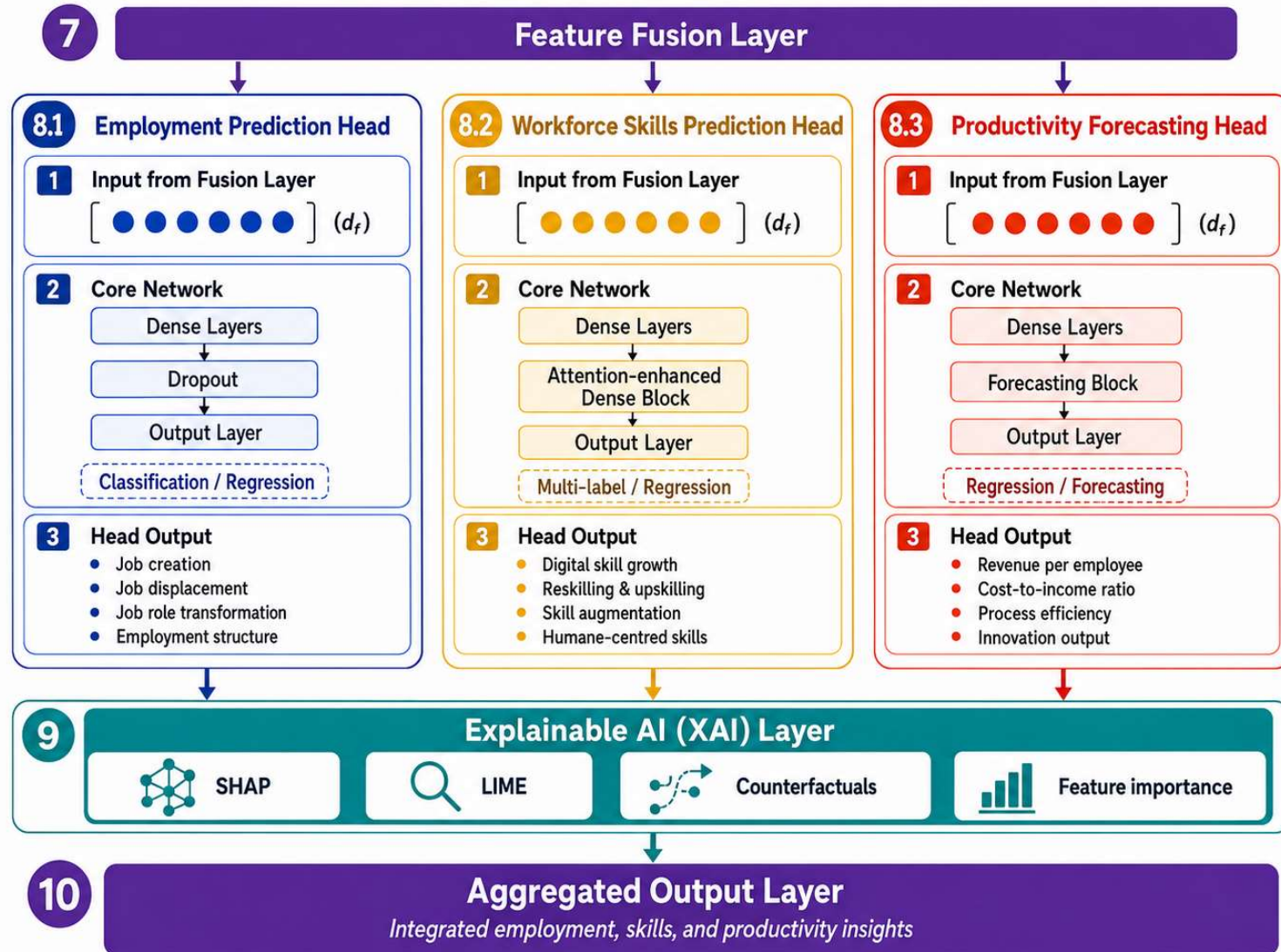
E. Outputs

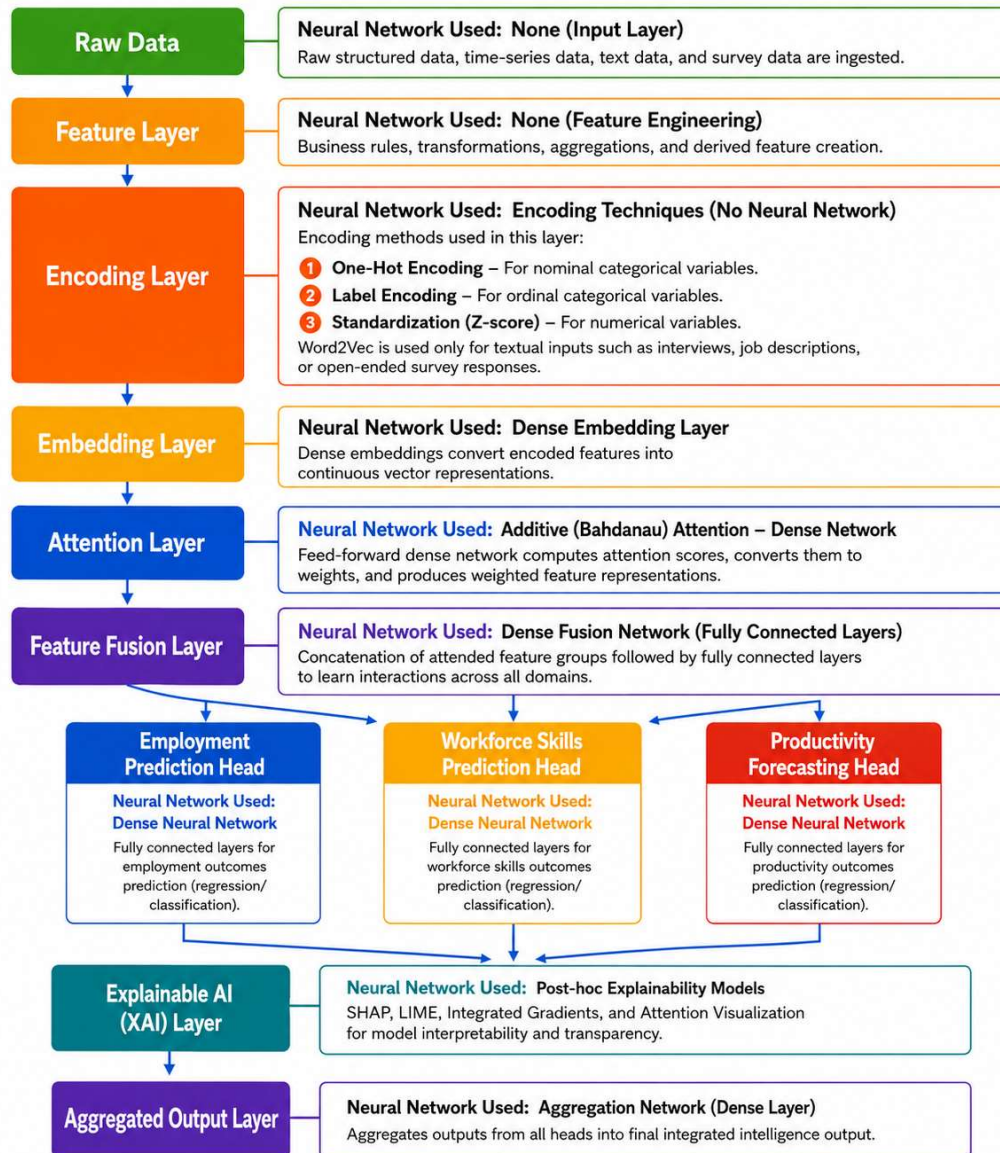
- prediction
- risk scoring
- explanations
- decision support

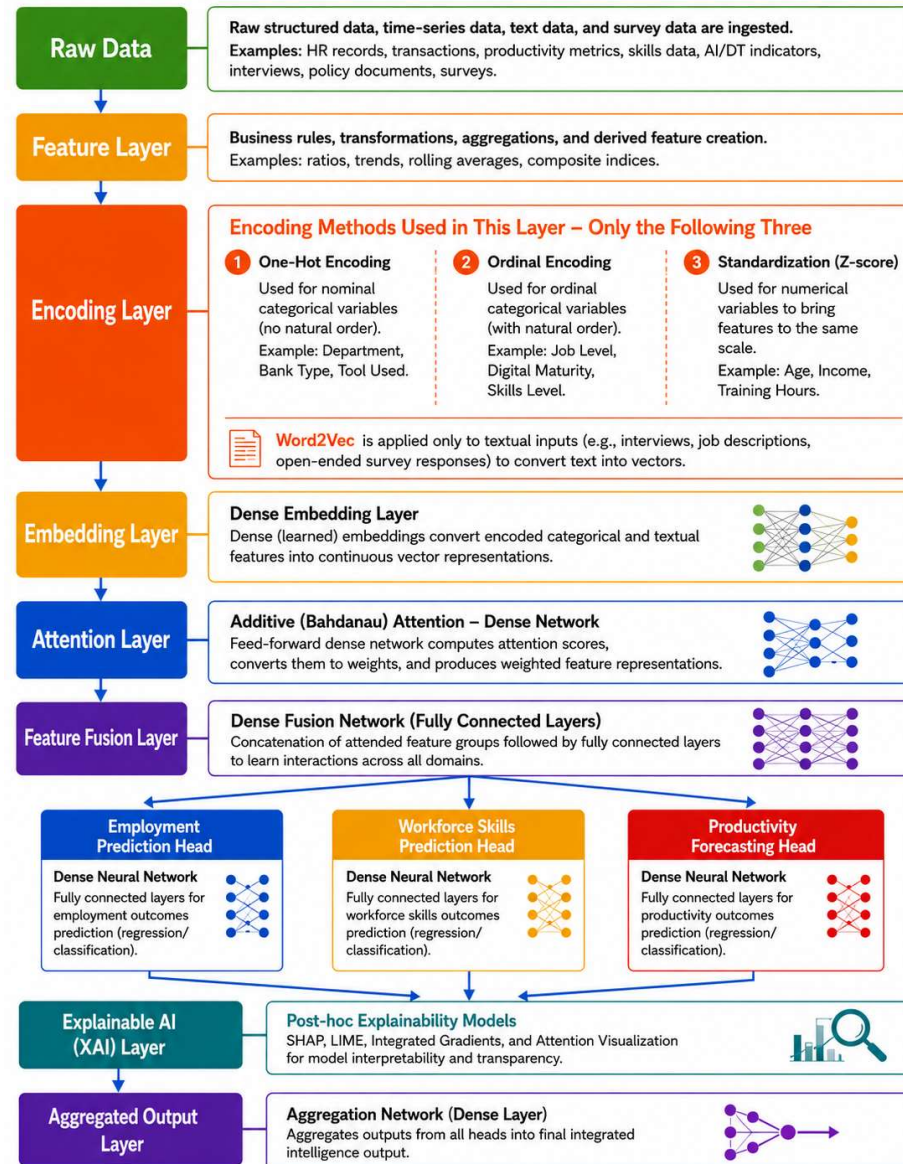


Outcome: fused features that combine multiple signals into richer, more predictive representations.

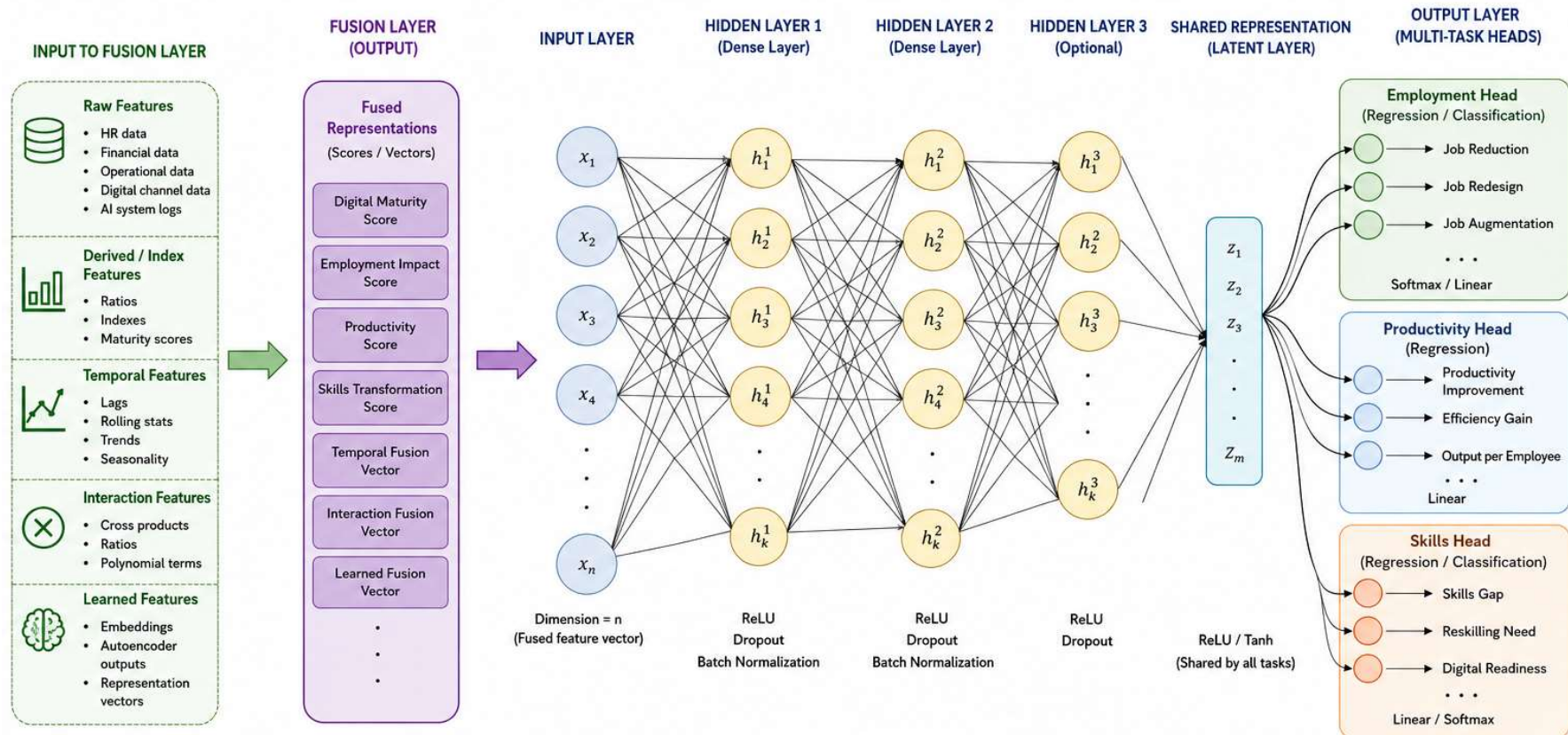
Prediction Heads Architecture







Typical Neural Network Architecture After Fusion Layer



ARCHITECTURE COMPONENTS

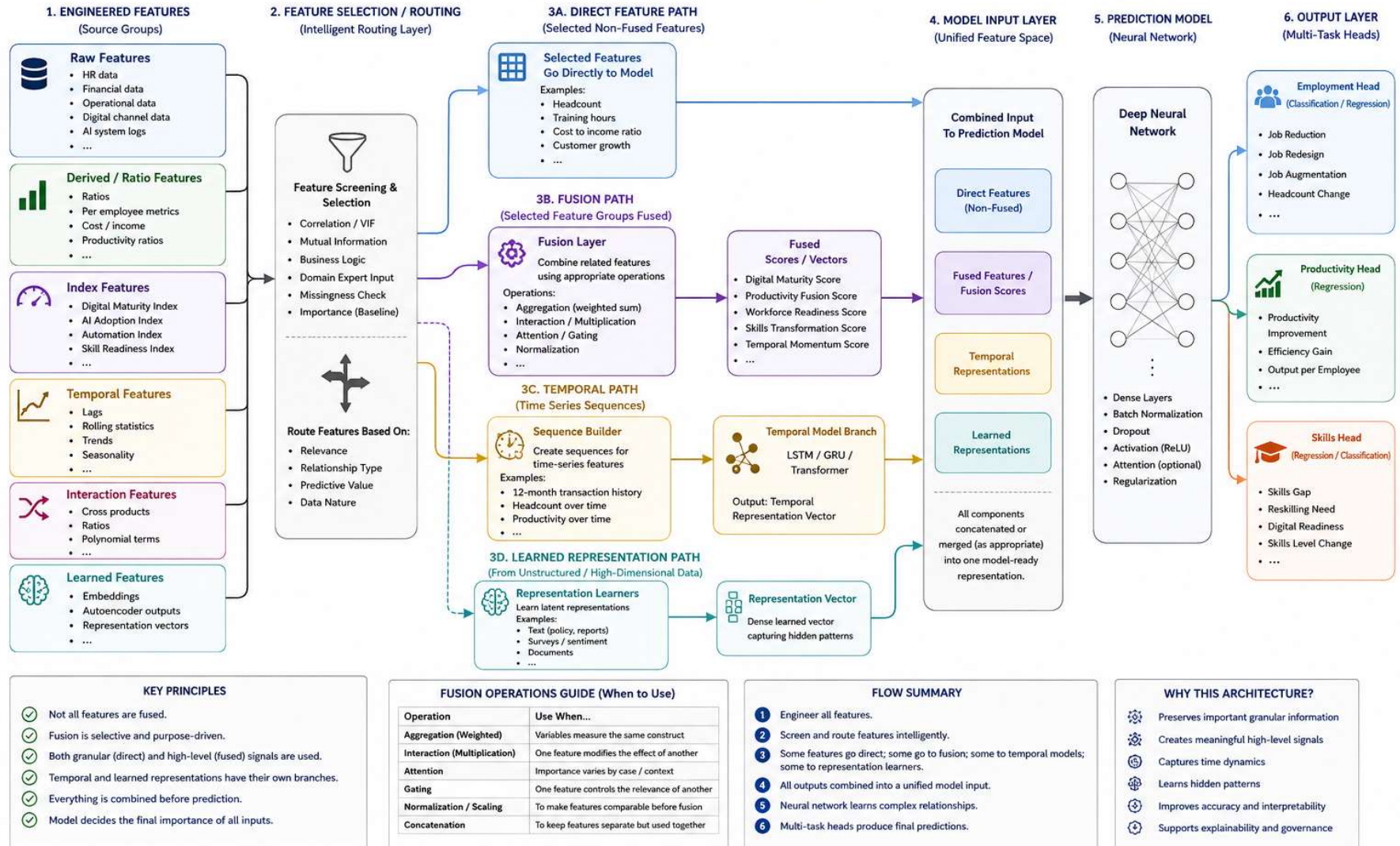
- Fully Connected Layers (Dense)
- Activation: ReLU (hidden), Linear / Softmax (output)
- Regularization: Dropout, L2, Batch Normalization
- Optimization: Adam / RMSprop
- Loss Functions: MSE, MAE, Cross-Entropy (task specific)

TYPICAL FLOW AFTER FUSION LAYER

- Fused features enter Input Layer
- Information flows through Hidden Layers
- Shared Latent Representation (captures complex patterns)
- Multi-task Output Heads produce predictions
- Loss is computed for each task and used to update the network

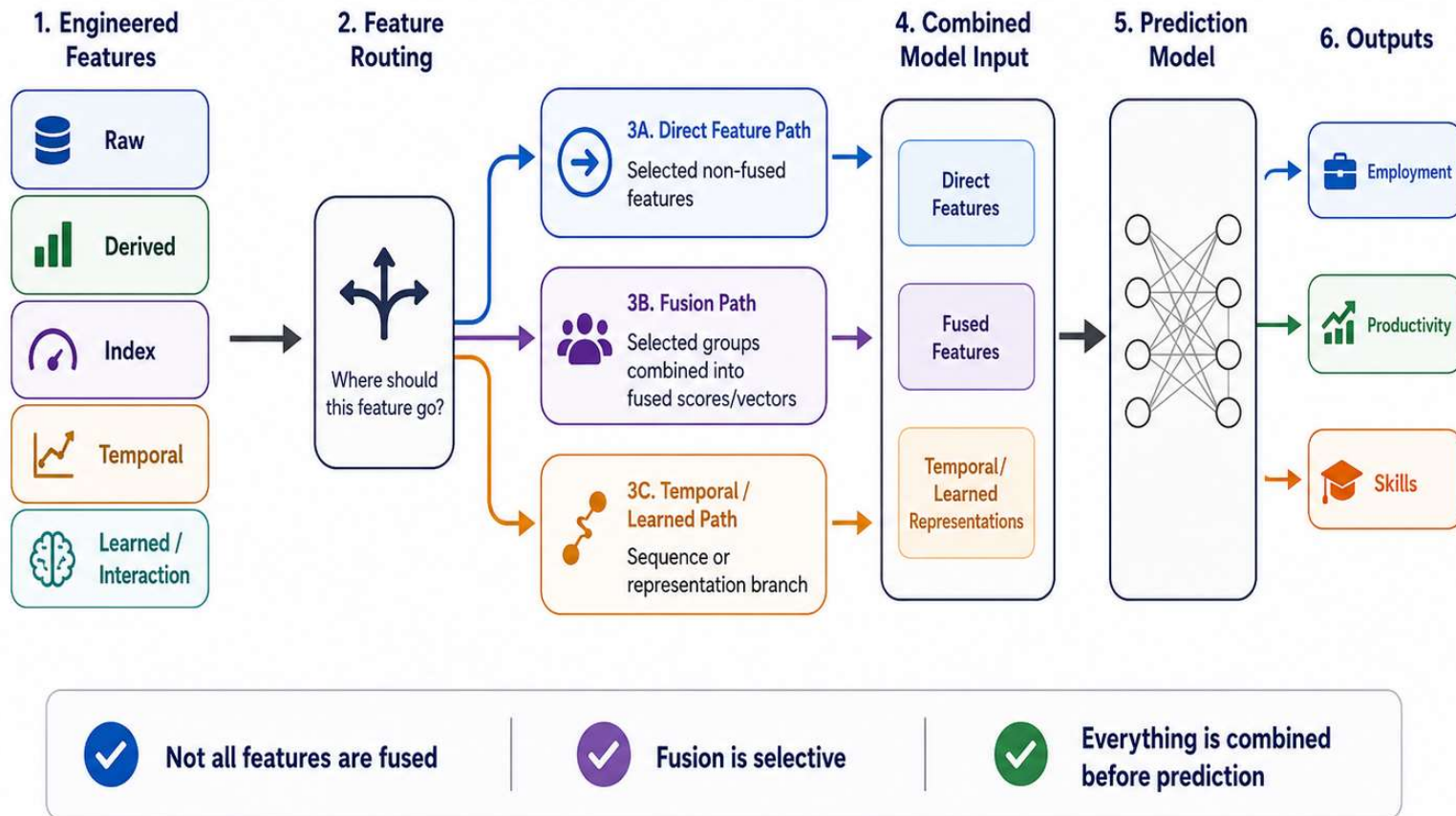
Routed Architecture: Not All Features Pass Through the Fusion Layer

Selected features are fused; others go directly to the model. All useful signals are combined before prediction.



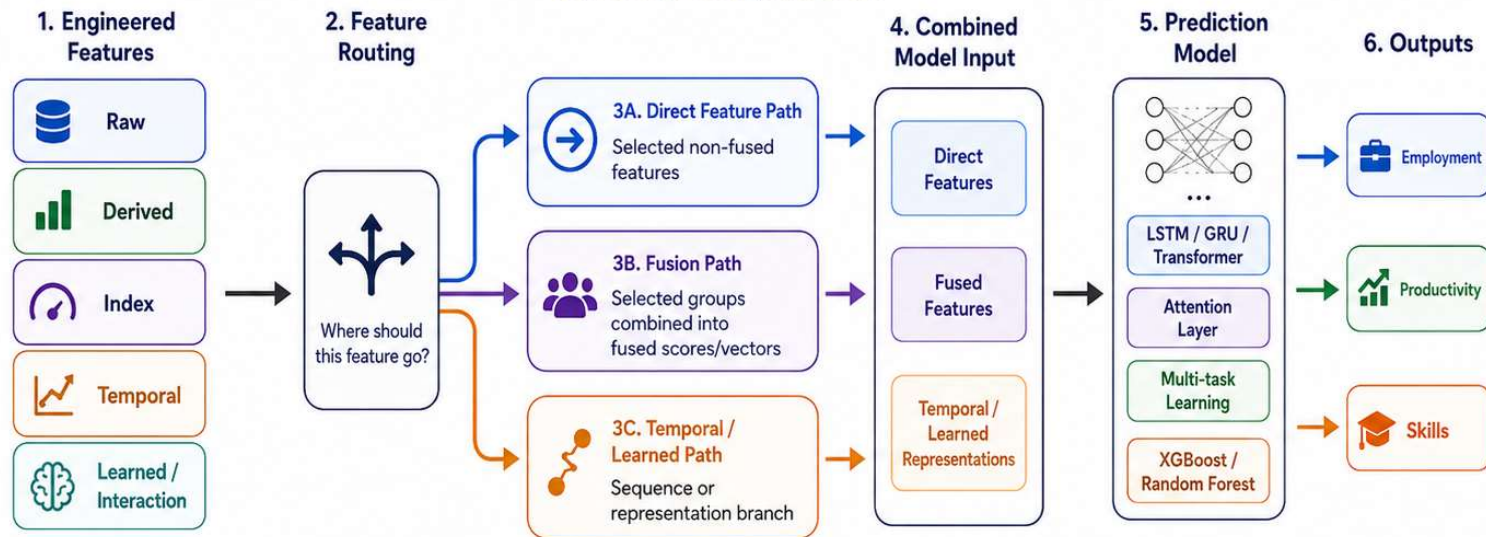
Simplified Routed Architecture After Feature Engineering

Some features go directly to the model; selected features go through fusion or specialized branches before prediction.



Simplified Routed Architecture After Feature Engineering

Some features go directly to the model; selected features go through fusion or specialized branches before prediction.



Not all features are fused



Fusion is selective



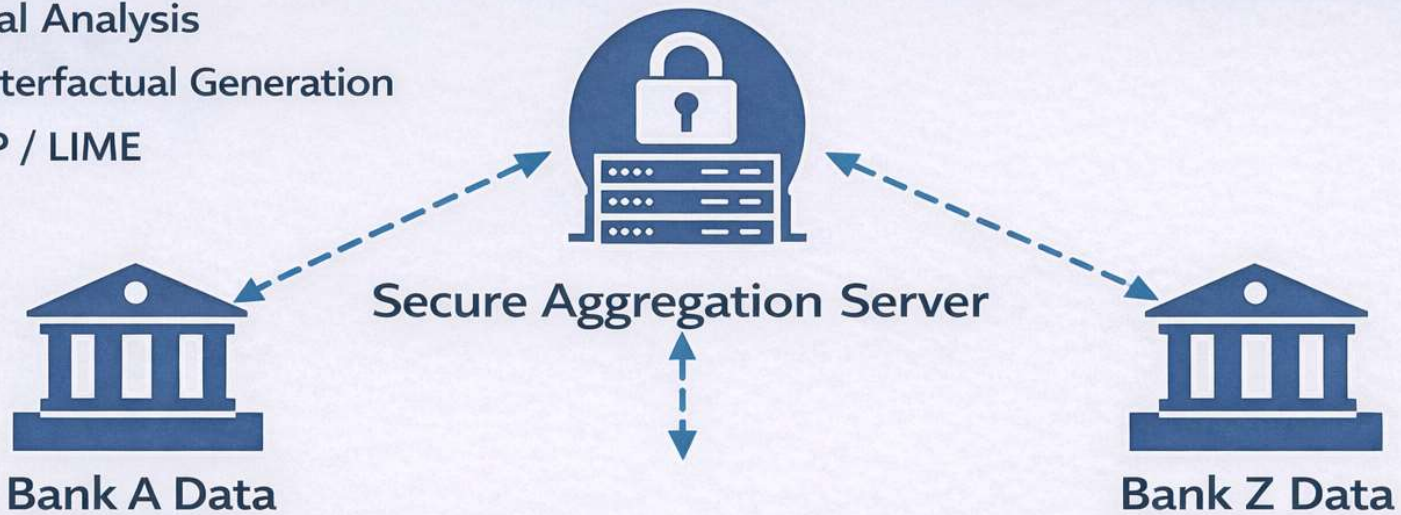
Everything is combined before prediction

Typical Model Components

Model component	Role
1. Dense neural network	Learns relationships across tabular banking features
2. LSTM / GRU / Transformer	Learns time-based change in employment, productivity and skills
3. Attention layer	Learns which feature groups matter most
4. Multi-task learning model	Predicts employment, productivity and skills together
5. XGBoost / Random Forest	Alternative or benchmark predictive models

Federated Learning & Privacy Layer

- Causal Analysis
- Counterfactual Generation
- SHAP / LIME



Raw data will be converted into raw features through a structured feature extraction and variable operationalisation process. Original records from operational systems, HR systems, financial reports, customer systems, surveys, and system logs will be cleaned, standardised, validated, and transformed into measurable variables. These raw features will represent the study's core constructs, including AI adoption, digital transformation, employment dynamics, productivity, and workforce skills. Feature selection will be guided by digital transformation theory, task-based labour economics, socio-technical systems theory, and the study's research objectives.

raw features are measurable variables extracted directly from original data sources before ratios, indexes, interactions, or temporal features are created.

Variable operationalisation = turning a research concept into measurable data.

Variable operationalisation will be used to convert the study's abstract constructs into measurable indicators and model-ready raw features. Constructs such as AI adoption, digital transformation, employment dynamics, productivity, and workforce skills will be translated into observable variables drawn from operational records, HR systems, AI system logs, surveys, interviews, and secondary datasets. This process ensures that the features used in the deep learning framework are theoretically grounded, empirically measurable, and aligned with the research objectives.

The study will begin with a broad pool of candidate raw variables drawn from HR, productivity, digital banking, AI system, financial, and operational datasets. These variables will be transformed into engineered features, including ratios, indices, temporal variables, interaction terms, and exposure measures. Feature shortlisting will then be conducted using theoretical relevance, data availability, missingness, multicollinearity checks, target-leakage screening, predictive contribution, and explainability analysis. The final explainable feature set will consist of the most influential and interpretable features for each outcome model: productivity, employment, workforce skills, and labour/operating costs.

Ratio features will be engineered by transforming raw operational, financial, HR, digital-channel, and AI-system variables into normalised measures of productivity, cost intensity, workforce structure, skills coverage, and digital adoption. Each ratio will be designed by defining a theoretically justified numerator and denominator, aligning both variables to the same time period and organisational unit, checking mathematical validity, screening for missing values and zero denominators, and testing for redundancy, target leakage, predictive value, and explainability. Ratio features will be retained only where they are conceptually meaningful, measurable across banks, interpretable to stakeholders, and relevant to the outcome model being estimated.

Interaction features will be engineered by combining key transformation drivers, such as the AI Adoption Index, Digital Transformation Index, Automation Intensity Index, Digital Transaction Share, and AI-Assisted Decision Share, with theoretically relevant moderating factors such as training hours, skill readiness, digital skills, routine job share, branch staff level, human–AI collaboration, data readiness, and workforce redeployment capacity. Each interaction feature will be designed to test whether the effect of AI adoption or digital transformation on productivity, employment, skills, and labour/operating costs becomes stronger, weaker, or different under specific workforce, organisational, operational, or technological conditions.

The interaction feature design process will involve identifying the main independent variable, selecting a conceptually justified moderating variable, aligning both variables to the same time period, organisational unit, and outcome model, and then constructing the interaction term through multiplication. Interaction terms will be screened for conceptual relevance, multicollinearity, redundancy, target leakage, data quality, predictive contribution, and explainability. Interaction features will be retained only where they are theoretically meaningful, measurable across banks, interpretable to stakeholders, and useful in explaining how AI and digital transformation effects vary across different workforce and organisational contexts.

Main driver × Moderating factor = Interaction feature

The feature tests whether AI adoption has a stronger effect on productivity, skills, employment, or labour costs when employee training is higher.

Learned features will be produced automatically by the deep learning model from raw input variables and engineered features. Unlike manually designed ratio, temporal, index, or interaction features, learned features are generated internally by neural network layers as hidden representations that capture complex, non-linear, and high-dimensional patterns in the data. These may include embeddings, temporal representations from LSTM or GRU layers, attention-weighted representations, latent workforce-change patterns, learned productivity patterns, and hidden skills-transition signals.

The production of learned features will involve feeding cleaned raw variables and engineered features into the deep learning architecture, where the model progressively transforms them through dense layers, recurrent layers, attention mechanisms, and shared representation layers. Through training, the model learns which combinations of variables are most useful for predicting productivity, employment, workforce skills, and labour/operating cost outcomes. Learned features will be used mainly inside the predictive model to improve forecasting accuracy, detect hidden relationships, capture non-linear effects, and identify complex patterns that may not be visible through manually engineered features alone. Their contribution will be interpreted using explainability methods such as SHAP, LIME, attention analysis, feature attribution, and sensitivity analysis.

The production of fusion features will involve aligning datasets by bank, department, branch, role, and time period, followed by combining related variables through feature concatenation, weighted indices, embedding fusion, attention-based fusion, or multi-task shared representation learning. Fusion features will allow the model to analyse how different dimensions of organisational transformation work together rather than in isolation. For example, AI adoption data may be fused with training data to understand workforce readiness, or digital transaction data may be fused with branch staffing data to understand productivity and employment effects. Fusion features will be used where the model needs to connect separate but related dimensions of the transformation process. They will support productivity prediction, skills-transition modelling, employment-impact forecasting, labour-cost analysis, scenario simulation, and explainable reporting. They will be retained where they improve predictive performance, reveal meaningful cross-system relationships, remain interpretable through explainability tools, and support the study's objective of forecasting the impact of AI and digital transformation on banking-sector outcomes.

Engineered features are manually designed. Learned features are discovered by the model. Fusion features are created by combining multiple data streams into one richer representation.

Learned features refer to latent representations automatically generated by machine learning or deep learning models from raw and derived features. Unlike manually engineered features such as ratios, indices, interaction terms, and temporal variables, learned features are not directly specified by the researcher. They are extracted from model layers, embeddings, hidden states, or representation-learning components and may be used to improve predictive accuracy, detect complex nonlinear patterns, and support downstream modelling. However, because they are less directly interpretable, their contribution will be examined using explainability techniques such as feature importance, SHAP values, attention analysis, or latent-space visualisation

Learned features are available, but they are usually internal model representations. They can be extracted and used, but they are not naturally visible or easily interpretable. They can be manipulated technically, but they should be interpreted through explainability methods rather than treated like ordinary variables

Learned features are not the same as model parameters. Parameters are the internal weights and biases learned during training, while learned features are the latent representations generated when input variables pass through those learned parameters. Parameters define how the model transforms the data; learned features are the transformed representations produced from that process.

In this study, raw data are first transformed into raw features, which serve as the foundational measurable variables. All subsequent engineered features, including ratio features, interaction features, index features, temporal features, learned features, and fusion features, are derived from these raw features either directly or through combinations, transformations, aggregations, or model-based representation learning.

Fusion features refer to integrated feature representations created by combining multiple categories of features into a unified model-ready representation. In this study, fusion features will be generated by combining raw features, derived features, temporal features, interaction features, ratio features, index features, and learned features to capture the combined effect of digital transformation, AI adoption, organisational capability, workforce skills, productivity, employment, and labour-cost dynamics.

Unlike individual features that represent a single variable or transformation, fusion features represent the joint information contained across different feature groups.

For example, a fusion feature may combine the Digital Transformation Index, AI Adoption Index, training intensity, productivity trends, employment changes, and learned latent representations into one integrated feature space.

This allows the model to analyse complex relationships that may not be visible when features are treated separately.

Fusion features are particularly important in this study because the effects of digital transformation and AI adoption are unlikely to operate through isolated variables. Their impact on productivity, employment, workforce skills, and labour costs is expected to emerge through the interaction of technological, organisational, temporal, and human-capability factors.

Fusion features therefore provide a mechanism for integrating these multiple dimensions into a stronger predictive and explanatory representation.

In the proposed modelling framework, fusion features will be created after raw, derived, temporal, interaction, ratio, index, and learned features have been generated. They will serve as the final integrated feature layer before prediction. This enables the model to use both human-engineered variables and model-generated latent representations to improve predictive accuracy, capture nonlinear relationships, and support explainability.

Feature fusion will be implemented as a researcher-guided and model-assisted process. The researcher will determine the candidate feature groups to be fused based on the theoretical framework, variable operationalisation table, data quality, and relevance to the study outcomes. These may include raw features, derived features, index features, ratio features, interaction features, temporal features, and learned representations. The fusion method will then be selected according to the modelling architecture, using approaches such as feature concatenation, weighted aggregation, dimensionality reduction, autoencoder-based fusion, attention-based fusion, or neural-network fusion layers. Where neural or attention-based fusion is used, the model will automatically learn the optimal internal weights and nonlinear combinations during training. The value of the fused representation will be validated through model comparison, ablation testing, predictive performance metrics, and explainability analysis.

Feature fusion will be guided by theoretical relevance, statistical validity, predictive contribution, multicollinearity checks, model performance, and explainability analysis.